# SPENDWISE

## A Personal Expense Tracker on a Full DevOps Pipeline

**DEVOPS FINAL CAPSTONE**

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Project Report • October 2026

FastAPI • PostgreSQL • React • Docker • GitHub Actions
Trivy • Terraform • Kubernetes • Helm • Prometheus • Grafana

# Contents

1. 1. Executive Summary
2. 2. Project Objectives and Scope
3. 3. Application Overview
4. 4. System Architecture and Delivery Flow
5. 5. Testing and Code Quality
6. 6. Containerisation and Local Stack
7. 7. Git, CI/CD and Image Provenance
8. 8. DevSecOps: Trivy and Vulnerability Remediation
9. 9. Infrastructure as Code with Terraform
10. 10. Kubernetes and Helm
11. 11. Observability: Prometheus and Grafana
12. 12. Live Demonstration Workflow
13. 13. Module Status and Limitations
14. 14. Conclusion and Future Improvements

Appendix A — Screenshot Evidence Gallery (42 images)

# 1. Executive Summary

SpendWise is a full-stack personal expense tracker developed as a DevOps final capstone. Users can record expenses with a title, amount, category, date and notes; filter expenses by category; and review totals and category-level spending on a dashboard.

The principal objective is to demonstrate a complete delivery and operations workflow around a deliberately simple application: automated testing, container builds, vulnerability scanning, image publishing, Kubernetes deployment, infrastructure-as-code, and observability.

The implementation combines a FastAPI backend, PostgreSQL database, React frontend, Docker, GitHub Actions, Trivy, Terraform, Kubernetes, Helm, Prometheus and Grafana.

# 2. Project Objectives and Scope

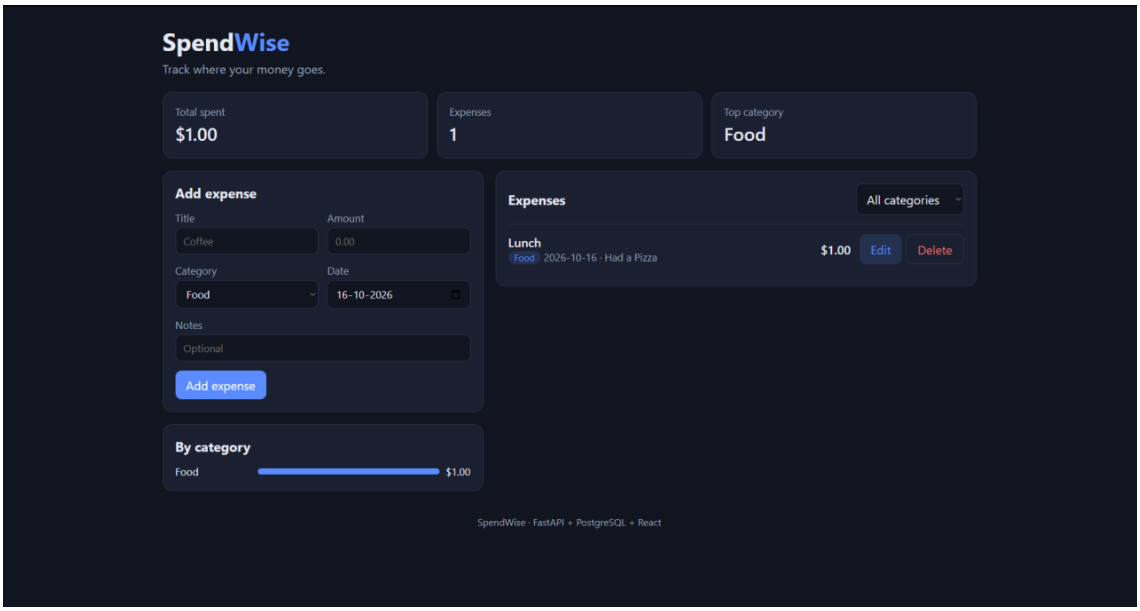
- Build a usable expense-tracking application with create, read, update and delete operations.
- Automate backend tests and linting, and build the frontend in continuous integration.
- Build container images using multi-stage Dockerfiles and non-root runtime users.
- Scan container images for known vulnerabilities and prevent publication when fixable HIGH or CRITICAL findings are detected.
- Publish images to GitHub Container Registry using commit-SHA tags for traceability.
- Deploy the application with Kubernetes and Helm, including health probes and ingress routing.
- Provision AWS network infrastructure through Terraform and document the EKS limitation.
- Expose application metrics and visualise service behaviour in Prometheus and Grafana.

# 3. Application Overview

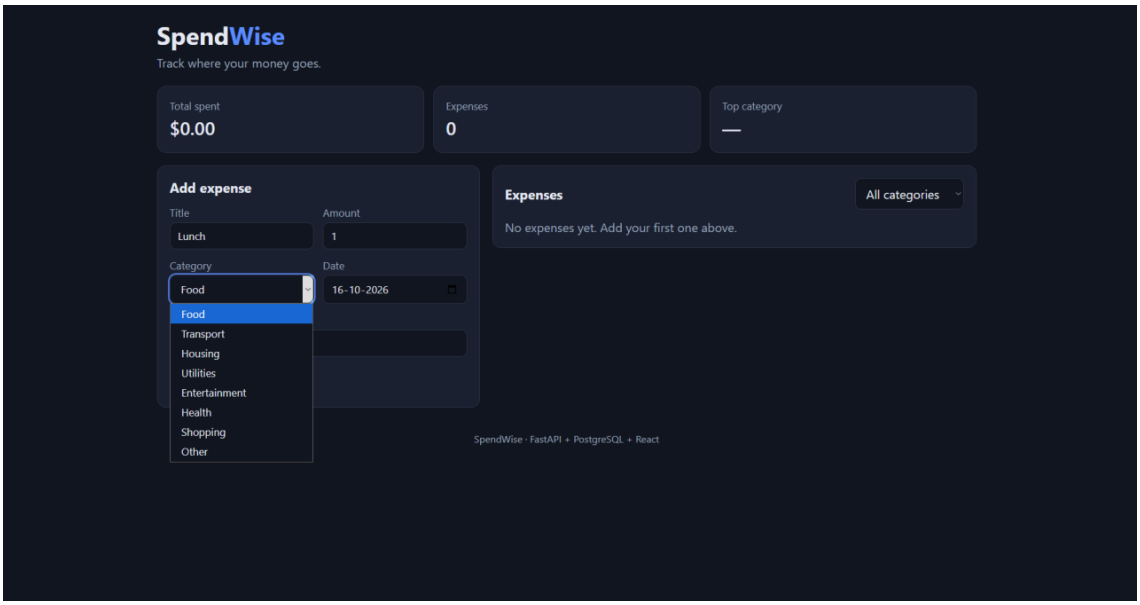
The application is intentionally small so the capstone can focus on engineering the delivery lifecycle. The frontend provides summary cards, an add/edit form, category filtering, category spending bars, a responsive layout and dark mode.

The FastAPI backend exposes CRUD endpoints under /api/expenses and a summary endpoint at /api/expenses/summary. The /health endpoint indicates that the process is alive, while /health/ready checks database reachability. PostgreSQL is accessed through SQLAlchemy 2, and Alembic migration 0001 creates the expenses table on startup. A PostgreSQL advisory lock is used to prevent multiple backend replicas from running migrations simultaneously.

Input validation includes positive amounts, fixed categories and field-length limits. Invalid input returns HTTP 422. The frontend is built with React 18 and Vite, then served by an unprivileged Nginx container that also forwards /api requests to the backend.




*Expense list and dashboard view*




*Add-expense form*

# 4. System Architecture and Delivery Flow

The workflow begins when a developer pushes code to GitHub. GitHub Actions runs backend linting and tests, builds the frontend, builds and scans backend and frontend images, and deploys to a temporary kind cluster for Helm-based smoke tests. Images are pushed to GitHub Container Registry and tagged with the full Git commit SHA.

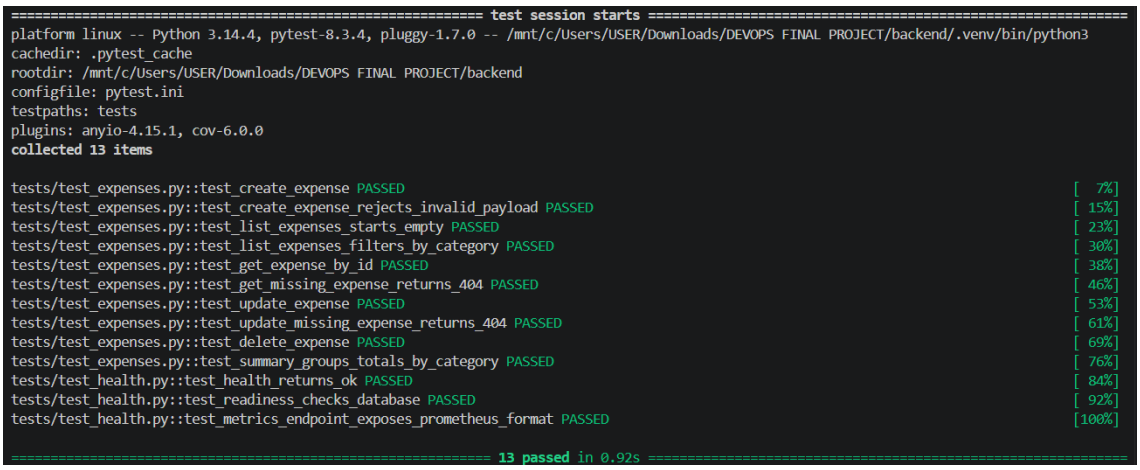
The Kubernetes deployment uses an Ingress to route / to the React frontend and /api to the FastAPI backend. The backend uses PostgreSQL. Prometheus discovers the backend metrics endpoint through a ServiceMonitor, and Grafana presents the metrics on the SpendWise API Overview dashboard.

Terraform defines the AWS network, including a VPC and public subnets. The supplied project notes state that the VPC and subnets were provisioned, while EKS was not applied because of account/free-plan constraints and the desire to avoid charges.

# 5. Testing and Code Quality

The project brief reports 13 automated pytest tests covering health, readiness, metrics, expense endpoints, invalid input and not-found cases. Tests use an in-memory SQLite database through a conftest.py dependency override, keeping tests isolated from the production database.

Ruff checks Python code style and common issues. Both linting and tests run in CI; a failure blocks the pipeline. The test evidence screenshot is included in the evidence gallery.




*Pytest test suite passing*

# 6. Containerisation and Local Stack

The backend Dockerfile uses a multi-stage build: the builder stage compiles Python wheels, while the runtime stage uses a slim Python image without compilers. The frontend Dockerfile builds the React app with Node and serves the output through unprivileged Nginx.

The project notes specify non-root runtime users (backend UID 10001 and frontend UID 101), OS package upgrades during the build, and built-in health checks. Docker Compose starts PostgreSQL 16, the backend and the frontend. The backend waits for the database, runs migrations and serves on port 8000; the frontend is available at http://localhost:3000.




*Docker Compose build evidence 1*



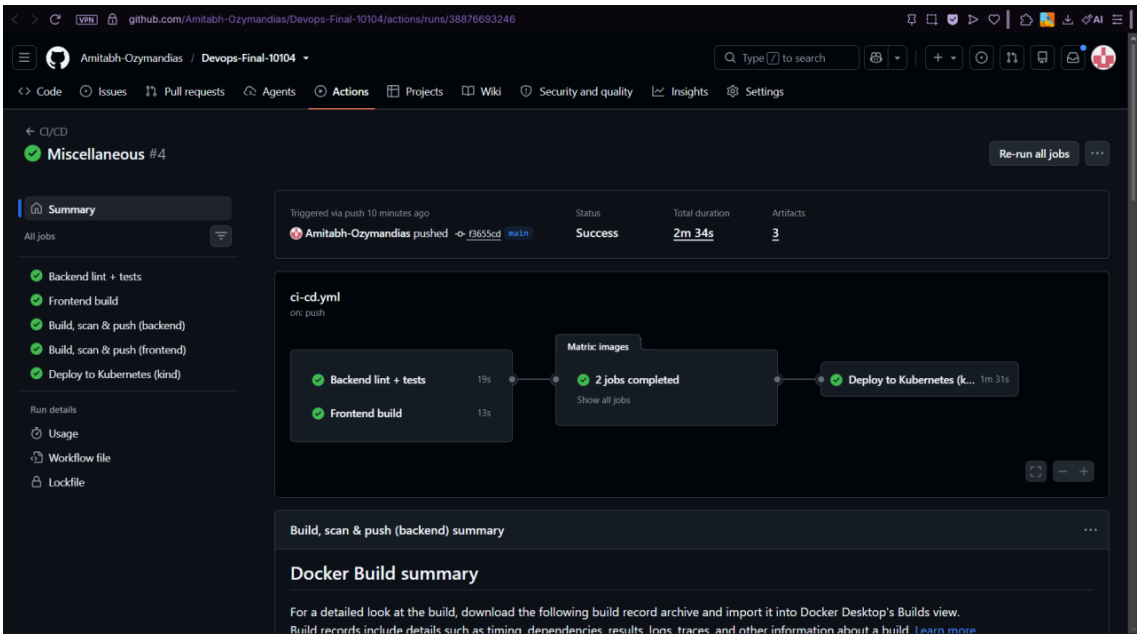

*Docker Compose build evidence 2*

# 7. Git, CI/CD and Image Provenance

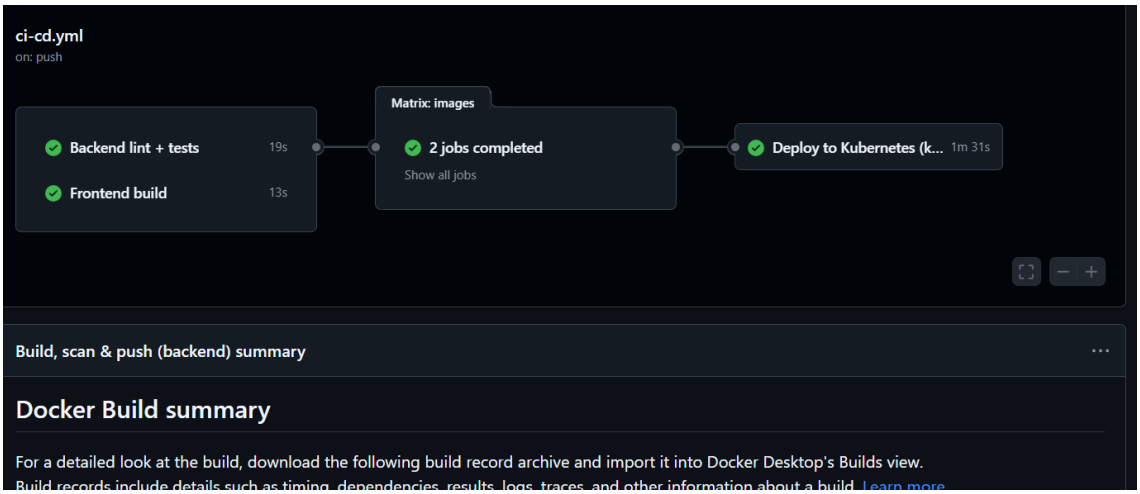
The public repository is github.com/Amitabh-Ozymandias/Devops-Final-10104. The supplied notes report more than 30 meaningful commits organised by module, including a dedicated security-fix commit. The .gitignore excludes environment files, Python caches, node_modules, virtual environments, Terraform state and terraform.tfvars.

The GitHub Actions workflow runs on pushes to main and contains five jobs: backend lint and tests; frontend build; backend image build, scan and push; frontend image build, scan and push; and deployment to a temporary kind cluster with Helm and Ingress smoke tests.

The images are published to ghcr.io/amitabh-ozymandias/spendwise-backend and ghcr.io/amitabh-ozymandias/spendwise-frontend. Each image is tagged with the full Git commit SHA rather than relying on the mutable latest tag. This makes deployments traceable to the source revision that produced them.




*Successful GitHub Actions workflow run*



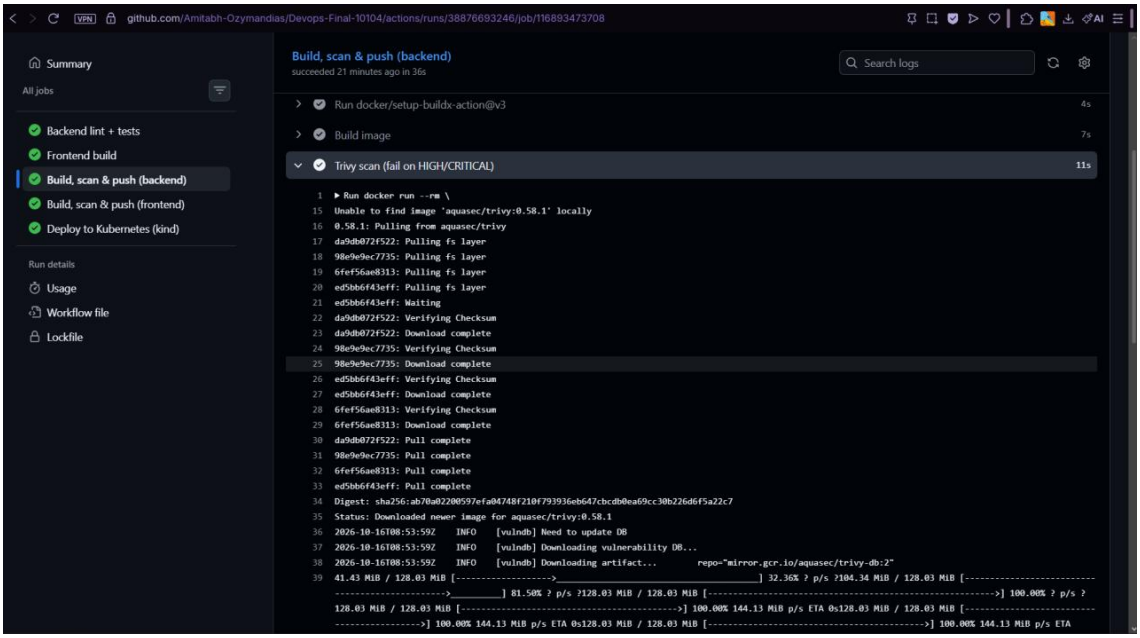

*CI/CD pipeline job graph*

# 8. DevSecOps: Trivy and Vulnerability Remediation

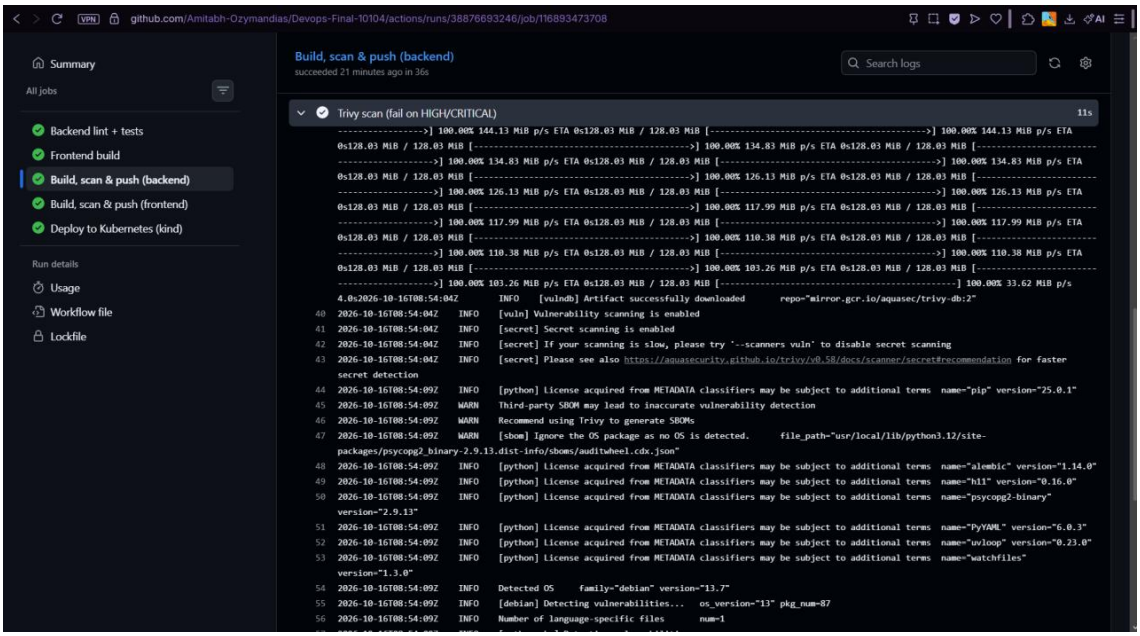
Trivy scans the backend and frontend images before publication. The scan covers operating-system packages and application dependencies, including Python packages and npm modules. The pipeline is configured to fail when a fix is available for a HIGH or CRITICAL vulnerability.

The project notes describe an initial scan that found three HIGH findings in Starlette 0.41.3, the web framework used beneath FastAPI. The notes identify CVE-2025-62727 as involving inefficient processing of crafted HTTP Range headers that could consume CPU and cause denial of service. The stated remediation was to upgrade to FastAPI 0.143.0 / Starlette 1.7.0, after which the next scan reported zero HIGH and zero CRITICAL findings.

This demonstrates a security gate in the delivery pipeline: an image with the reported vulnerable dependency was blocked from publication until the dependency was updated.




*Trivy backend image scan start*




*Trivy backend image scanning output*

# 9. Infrastructure as Code with Terraform

Terraform defines AWS resources using the official terraform-aws-modules. The described network includes a VPC with CIDR 10.20.0.0/16, two public subnets across availability zones, an internet gateway and route tables. The configuration also contains optional EKS cluster and managed node group resources controlled by create_eks.

The reported plan contained 11 resources to add, zero to change and zero to destroy. The public subnets were ap-southeast-2a (10.20.0.0/24) and ap-southeast-2b (10.20.1.0/24), with tags for Kubernetes load balancers. The project notes state that apply created 11 resources and destroy subsequently removed 11 resources.

The AWS account constraints described in the supplied material prevented EKS from being applied. Therefore, the report treats EKS as defined in code but not provisioned, and does not represent the AWS deployment as a live EKS cluster. The notes state that nothing remains running in AWS, avoiding ongoing cost.

```
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terraform$ cd "/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terra
form"
sed -i 's/^aws_region.*/aws_region = "ap-southeast-2"/' terraform.tfvars
terraform fmt                    # tidies the spacing in terraform.tfvars
grep -E "aws_region|create_eks" terraform.tfvars   # aws_region = "ap-southeast-2", create_eks = false
terraform validate               # Success! The configuration is valid.
terraform plan                   # ☁️ should end: "Plan: 9 to add, 0 to change, 0 to destroy"
terraform.tfvars
aws_region   = "ap-southeast-2"
create_eks         = false
Success! The configuration is valid.

data.aws_availability_zones.available: Reading...
data.aws_availability_zones.available: Read complete after 2s [id=ap-southeast-2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # module.vpc.aws_default_network_acl.this[0] will be created
  + resource "aws_default_network_acl" "this" {
      + arn                    = (known after apply)
      + default_network_acl_id = (known after apply)
      + id                     = (known after apply)
      + owner_id               = (known after apply)
      + tags                   = {
          + "Name" = "spendwise-vpc-default"
        }
      + tags_all               = {
          + "Environment" = "dev"
          + "ManagedBy"   = "terraform"
          + "Name"        = "spendwise-vpc-default"
          + "Project"     = "spendwise"
        }
      + vpc_id                 = (known after apply)
```

*Terraform validation and plan*

```
      + egress {
          + action          = "allow"
          + from_port       = 0
          + ipv6_cidr_block = "::/0"
          + protocol        = "-1"
          + rule_no         = 101
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }
      + egress {
          + action          = "allow"
          + cidr_block      = "0.0.0.0/0"
          + from_port       = 0
          + protocol        = "-1"
          + rule_no         = 100
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }

      + ingress {
          + action          = "allow"
          + from_port       = 0
          + ipv6_cidr_block = "::/0"
          + protocol        = "-1"
          + rule_no         = 101
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }
      + ingress {
          + action          = "allow"
          + cidr_block      = "0.0.0.0/0"
          + from_port       = 0
          + protocol        = "-1"
          + rule_no         = 100
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }
```

*Terraform plan evidence 2*

```
  # module.vpc.aws_default_route_table.default[0] will be created
  + resource "aws_default_route_table" "default" {
      + arn                    = (known after apply)
      + default_route_table_id = (known after apply)
      + id                     = (known after apply)
      + owner_id               = (known after apply)
      + route                  = (known after apply)
      + tags                   = {
          + "Name" = "spendwise-vpc-default"
        }
      + tags_all               = {
          + "Environment" = "dev"
          + "ManagedBy"   = "terraform"
          + "Name"        = "spendwise-vpc-default"
          + "Project"     = "spendwise"
        }
      + vpc_id                 = (known after apply)

      + timeouts {
          + create = "5m"
          + update = "5m"
        }
    }

  # module.vpc.aws_default_security_group.this[0] will be created
  + resource "aws_default_security_group" "this" {
      + arn                    = (known after apply)
      + description            = (known after apply)
      + egress                 = (known after apply)
      + id                     = (known after apply)
      + ingress                = (known after apply)
      + name                   = (known after apply)
      + name_prefix            = (known after apply)
      + owner_id               = (known after apply)
      + revoke_rules_on_delete = false
      + tags                   = {
```

*Terraform plan evidence 3*

# 10. Kubernetes and Helm

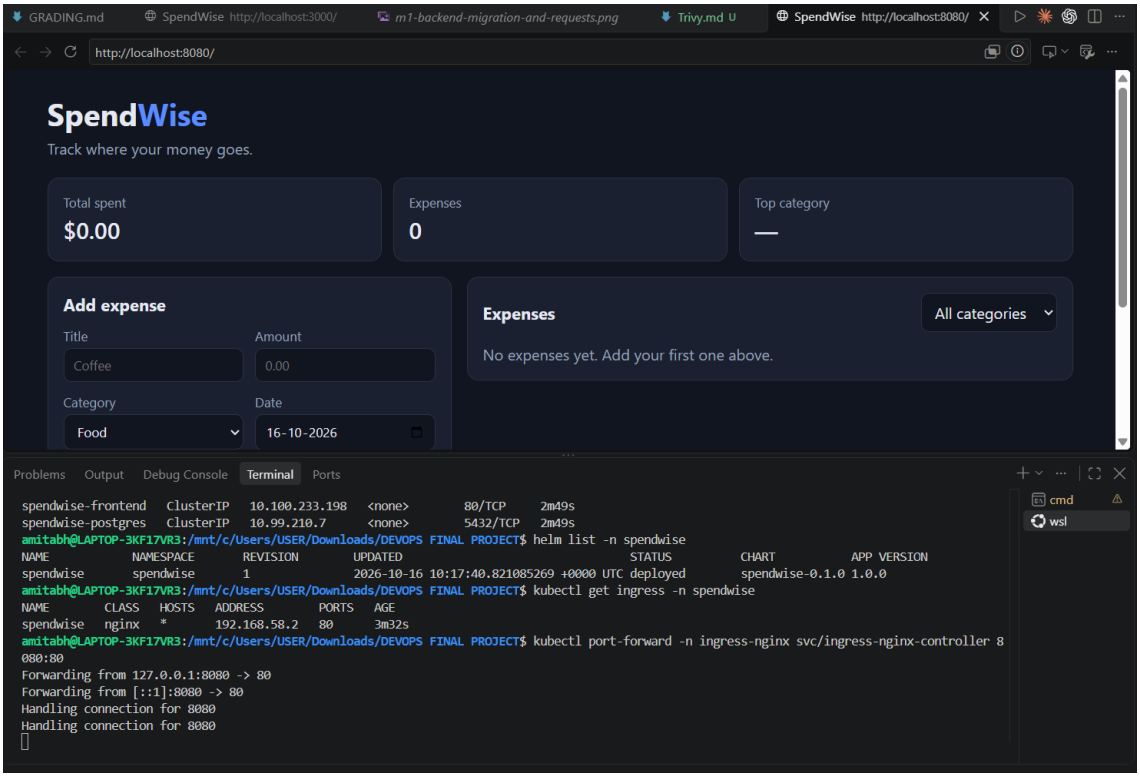
The Kubernetes configuration creates spendwise and monitoring namespaces. The Helm chart deploys a backend Deployment with two replicas and startup, liveness and readiness probes; a frontend Deployment with two replicas; a PostgreSQL StatefulSet with credentials stored in a Secret; ClusterIP Services; and an Ingress.

Ingress routes / to the frontend and /api to the backend. The project was deployed on minikube using helm upgrade --install, and the supplied notes report that all pods were Running. Pods run as non-root and drop Linux capabilities.

```
Problems  Output  Debug Console  Terminal  Ports

amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$ kubectl get pods -n spendwise -w
NAME                                  READY   STATUS    RESTARTS   AGE
spendwise-backend-544b474b4d-2nnn7    1/1     Running   0          2m16s
spendwise-backend-544b474b4d-4fcvt    1/1     Running   0          2m16s
spendwise-frontend-f76c78855-ms6wt    1/1     Running   0          2m16s
spendwise-frontend-f76c78855-t6cxd    1/1     Running   0          2m16s
spendwise-postgres-0                  1/1     Running   0          2m16s
^Camitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$ kubectl get pods -n spendwise
NAME                                  READY   STATUS    RESTARTS   AGE
spendwise-backend-544b474b4d-2nnn7    1/1     Running   0          2m40s
spendwise-backend-544b474b4d-4fcvt    1/1     Running   0          2m40s
spendwise-frontend-f76c78855-ms6wt    1/1     Running   0          2m40s
spendwise-frontend-f76c78855-t6cxd    1/1     Running   0          2m40s
spendwise-postgres-0                  1/1     Running   0          2m40s
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$ kubectl get svc -n spendwise
NAME                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
spendwise-backend   ClusterIP   10.108.111.52    <none>        8000/TCP   2m49s
spendwise-frontend  ClusterIP   10.100.233.198   <none>        80/TCP     2m49s
spendwise-postgres  ClusterIP   10.99.210.7      <none>        5432/TCP   2m49s
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$ helm list -n spendwise
NAME            NAMESPACE       REVISION        UPDATED                                 STATUS          CHART           APP VERSION
spendwise       spendwise       1               2026-10-16 10:17:40.821085269 +0000 UTC deployed        spendwise-0.1.0 1.0.0
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$ kubectl get ingress -n spendwise
NAME        CLASS   HOSTS   ADDRESS        PORTS   AGE
spendwise   nginx   *       192.168.58.2   80      3m32s
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$ kubectl port-forward -n ingress-nginx svc/ingress-nginx-controller 8
080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
```

*Kubernetes pods, services, Helm and Ingress*




*SpendWise application served through Ingress*

# 11. Observability: Prometheus and Grafana

The backend exposes /metrics in Prometheus format. The documented metrics include http_requests_total, labelled by endpoint, method and status; http_request_duration_seconds for request duration; and spendwise_expenses_created_total as a business metric.

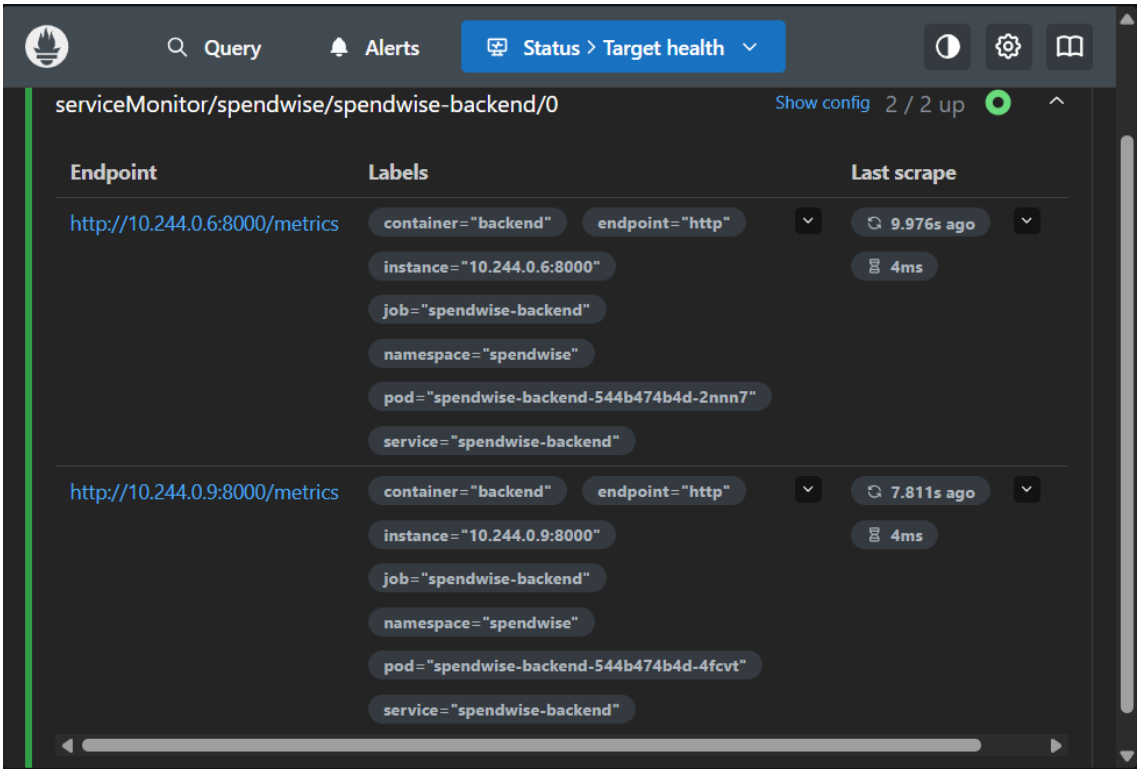
Prometheus uses kube-prometheus-stack and discovers the application through a ServiceMonitor. The supplied evidence shows serviceMonitor/spendwise/spendwise-backend/0 with two of two targets UP, corresponding to the two backend pods. The query up{namespace="spendwise"} returns 1 for each healthy scraped backend target.

The documented request-rate query is sum by (handler, method) (rate(http_requests_total{namespace="spendwise"}[1m])). The Grafana dashboard visualises requests per second, 5xx error rate, p95 latency, backend pods up, requests by endpoint, responses by status, p50/p95/p99 latency and expenses created per category.

The screenshots show a short burst of endpoint traffic and category metric activity during the captured interval. These are point-in-time observations, not a long-term performance benchmark. The dashboard's error-rate panel shows “No data” in the supplied view, so no numerical 5xx rate is asserted.

```
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$ kubectl port-forward -n spendwise svc/spendwise-backend 8000:8000 &
sleep 2; curl -s localhost:8000/metrics | grep -E "^http_requests_total|^spendwise" | head -20
kill %1
[1] 114030
Forwarding from 127.0.0.1:8000 -> 8000
Forwarding from [::1]:8000 -> 8000
Handling connection for 8000
spendwise_expenses_created_total{category="Food"} 137.0
spendwise_expenses_created_created{category="Food"} 1.7921520829135823e+09
http_requests_total{handler="/api/expenses/summary",method="GET",status="2xx"} 1.0
http_requests_total{handler="/api/expenses",method="POST",status="2xx"} 137.0
http_requests_total{handler="/api/expenses/{expense_id}",method="GET",status="4xx"} 161.0
http_requests_total{handler="/api/expenses",method="GET",status="2xx"} 154.0
[1]+  Terminated              kubectl port-forward -n spendwise svc/spendwise-backend 8000:8000
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT$
```

*Backend Prometheus metrics endpoint*




*Prometheus targets for both SpendWise backend pods*

# 12. Live Demonstration Workflow

15. The application initially displays the subtitle “Track where your money goes.”
16. The subtitle is changed in frontend/src/App.jsx, then committed and pushed to main.
17. GitHub Actions starts automatically and the five jobs are expected to pass.
18. New images are published to GHCR with the new commit SHA.
19. helm upgrade rolls out the new version and replaces the old pods.
20. The deployments run the new image SHA.
21. The application displays the updated subtitle.

The supplied image set contains evidence for the application, build, security, infrastructure, Kubernetes and monitoring stages. The seven demo-specific screenshots mentioned in the slide text were not present as separate files in the provided 42-image set, so the sequence above is documented from the supplied project description rather than presented as additional screenshot evidence.

# 13. Module Status and Limitations

| Module | Area | Reported status / qualification |
|---|---|---|
| **M1** | Application, frontend, backend and database | Complete |
| **M2** | Testing and code quality | Complete |
| **M3** | Git and GitHub | Complete; commit-history image not in supplied image set |
| **M4** | Docker and Compose | Complete |
| **M5** | GitHub Actions CI/CD | Complete |
| **M6** | DevSecOps / Trivy | Complete |
| **M7** | Terraform / AWS | VPC and subnets provisioned; EKS code not applied |
| **M8** | Kubernetes and Helm | Complete on minikube |
| **M9** | Prometheus and Grafana | Complete |
| **M10** | Documentation and demo | Complete per project notes; demo-specific screenshots not in the supplied image set |

The principal deployment limitation is that EKS was not applied to AWS. Kubernetes deployment evidence instead refers to minikube and the CI pipeline's temporary kind cluster. The report preserves that distinction.

# 14. Conclusion and Future Improvements

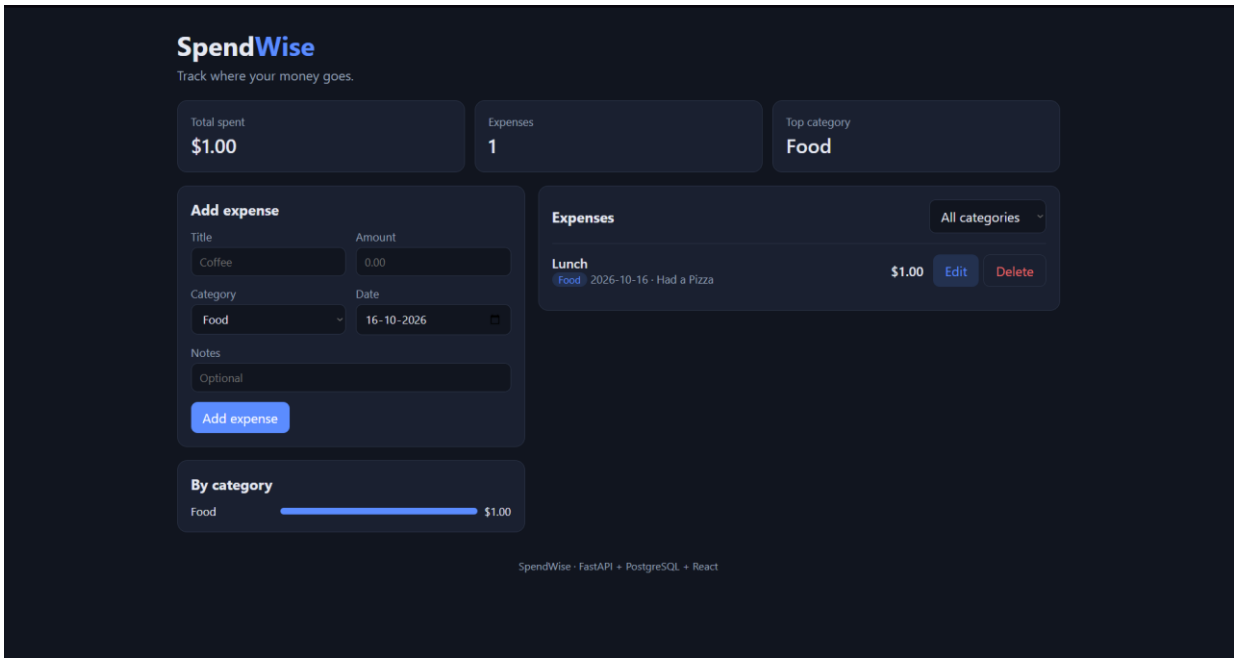
SpendWise demonstrates an end-to-end DevOps workflow around a full-stack application. Automated tests and linting provide a quality gate; multi-stage, non-root containers support safer packaging; Trivy introduces vulnerability scanning; SHA-tagged GHCR images improve traceability; Helm and Kubernetes handle deployment; Terraform describes cloud infrastructure; and Prometheus plus Grafana provide operational visibility.

- Add an authenticated production deployment environment and a managed secret store.
- Introduce explicit alert rules for elevated 5xx rate, latency and unavailable replicas.
- Retain longer monitoring windows and run load tests before drawing performance conclusions.
- Add database backup and restore validation, plus a documented recovery procedure.
- Apply the EKS configuration only when account permissions, region constraints and cost controls permit it.
- Include the dedicated Git history and seven live-demo screenshots in a future evidence revision if they are required for assessment.

# Appendix A — Screenshot Evidence Gallery

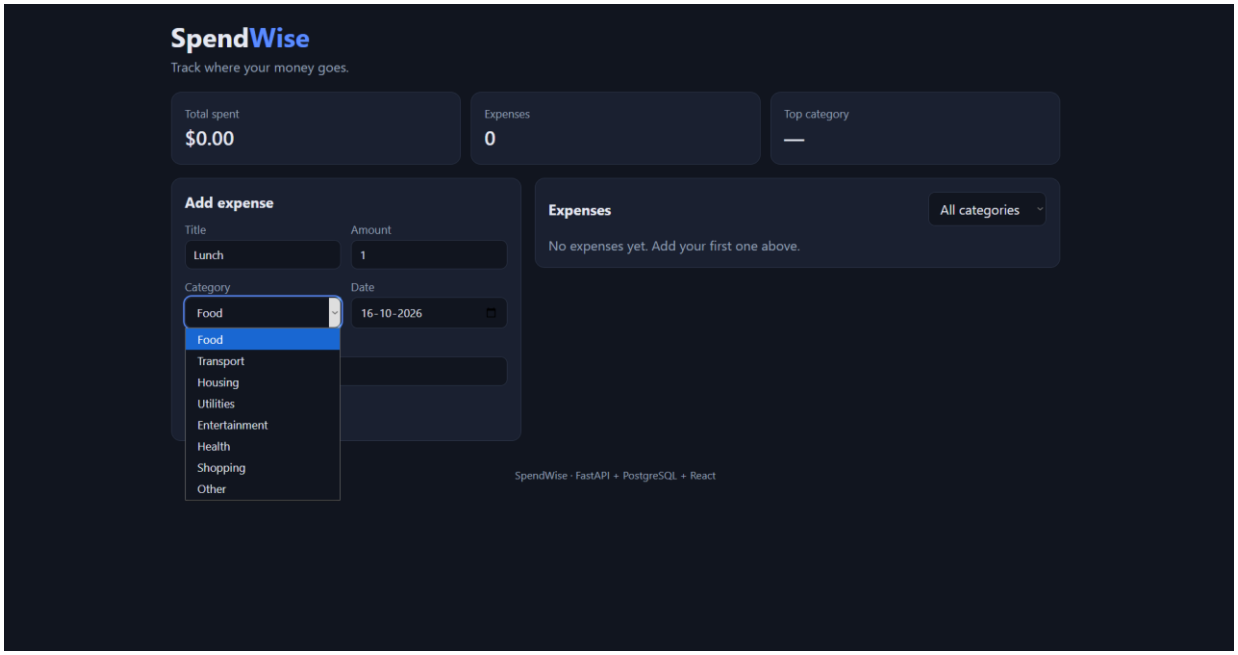
All 42 screenshot files supplied with the project brief are included below. Captions describe the evidence represented by each filename; they do not claim results beyond the supplied project notes and screenshots.

## A01. Expense list and dashboard view




Evidence group: Application | File: m1-app-expense-list(2).png

## A02. Add-expense form



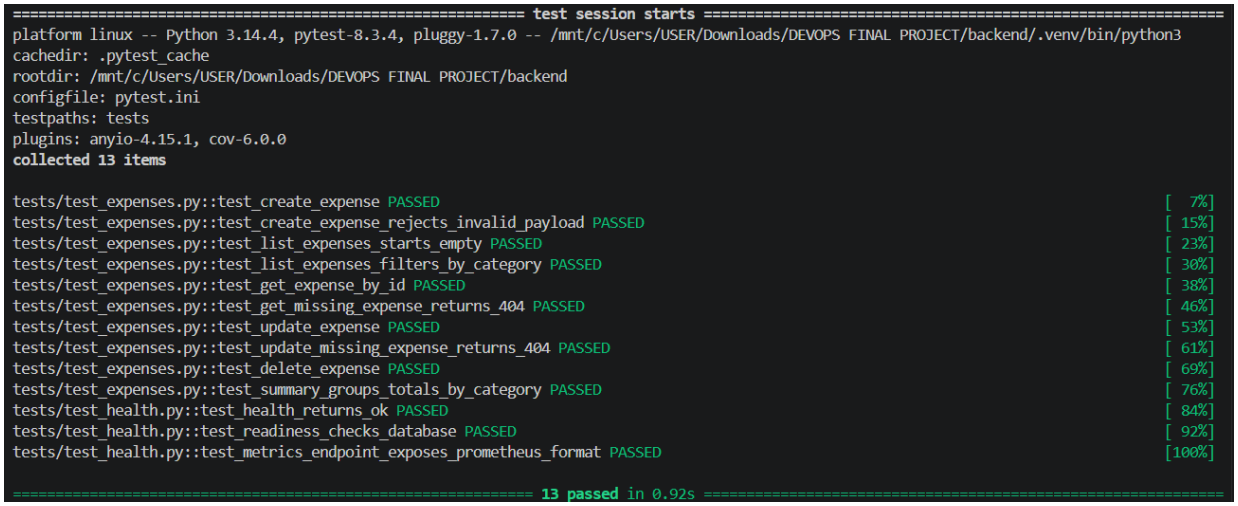

Evidence group: Application | File: m1-app-add-expense(2).png

## A03. Backend migration and API request evidence

```
backend-1   | INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
backend-1   | INFO  [alembic.runtime.migration] Will assume transactional DDL.
backend-1   | INFO  [alembic.runtime.migration] Running upgrade  -> 0001, create expenses table
backend-1   | INFO:     Started server process [1]
backend-1   | INFO:     Waiting for application startup.
backend-1   | INFO:     Application startup complete.
backend-1   | INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
backend-1   | INFO:     127.0.0.1:52134 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:51516 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:51358 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:48172 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:59824 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:39782 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:46834 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:43340 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:49842 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:39038 - "GET /health HTTP/1.1" 200 OK
postgres-1  | 2026-10-16 07:50:39.721 UTC [61] LOG:  checkpoint starting: time
postgres-1  | 2026-10-16 07:50:49.800 UTC [61] LOG:  checkpoint complete: wrote 103 buffers (0.6%); 0 WAL file(s) added, 0 removed, 0 recycled
; write=10.051 s, sync=0.017 s, total=10.079 s; sync files=57, longest=0.003 s, average=0.001 s; distance=466 kB, estimate=466 kB; lsn=0/199A7
D8, redo lsn=0/199A7A0
backend-1   | INFO:     127.0.0.1:57668 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:54172 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:47754 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:50090 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:47412 - "GET /health HTTP/1.1" 200 OK
frontend-1  | 172.18.0.1 - - [16/Oct/2026:07:53:04 +0000] "GET / HTTP/1.1" 200 477 "-" "curl/8.9.0" "-"
backend-1   | INFO:     172.18.0.4:34174 - "GET /api/expenses/summary HTTP/1.0" 200 OK
frontend-1  | 172.18.0.1 - - [16/Oct/2026:07:53:04 +0000] "GET /api/expenses/summary HTTP/1.1" 200 40 "-" "curl/8.9.0" "-"
backend-1   | INFO:     172.18.0.1:37914 - "GET /health HTTP/1.1" 200 OK
backend-1   | INFO:     127.0.0.1:53286 - "GET /health HTTP/1.1" 200 OK
```

Evidence group: Application | File: m1-backend-migration-and-requests(2).png

## A04. Pytest test suite passing




Evidence group: Testing | File: m2-pytest-passing(2).png

## A05. Docker Compose build evidence 1



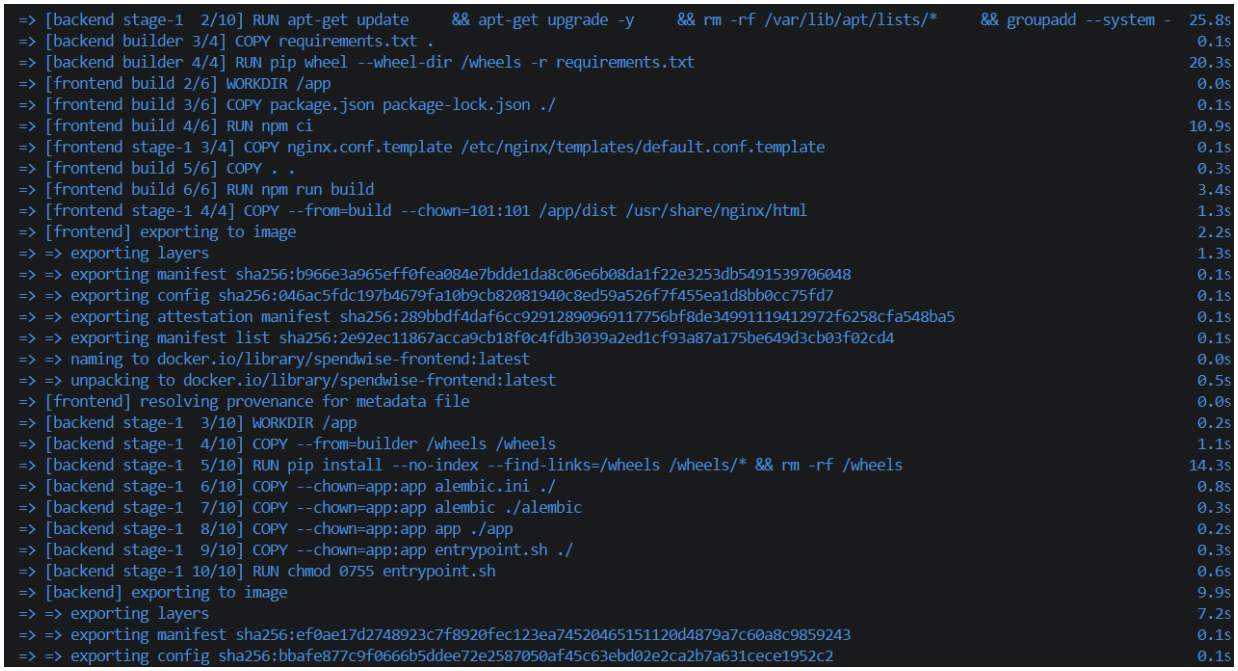

Evidence group: Docker and Compose | File: m4-compose-build-1(2).png

## A06. Docker Compose build evidence 2

```
=> => sha256:8f7ead5f2dc0fb85f6d6b3c81284ccd4b80b783030d202a18ba5ccc5ee98f451 403B / 403B                                  0.3s
=> => sha256:048c29a613179205d158f722a495c6b3bbd77439864c075ea5db2d42331d1dcc 956B / 956B                                  0.3s
=> => sha256:513bae533eb237e219e2d3412e0dbb9cd94c25b2b84530b5131fda9512c0e773 627B / 627B                                  0.3s
=> => sha256:945f5aaec3d7649d815440e6030ec4e4f562488ee50fa779a69f920cab3c5871 2.80kB / 2.80kB                              0.6s
=> => sha256:576fe098e8467d48ff489eb63f411a74b9e3769957a5c26462c4468a731c3e3d 1.79MB / 1.79MB                              1.1s
=> => sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB                              1.7s
=> => extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870                                   0.2s
=> => extracting sha256:576fe098e8467d48ff489eb63f411a74b9e3769957a5c26462c4468a731c3e3d                                   0.2s
=> => extracting sha256:945f5aaec3d7649d815440e6030ec4e4f562488ee50fa779a69f920cab3c5871                                   0.0s
=> => extracting sha256:513bae533eb237e219e2d3412e0dbb9cd94c25b2b84530b5131fda9512c0e773                                   0.0s
=> => extracting sha256:048c29a613179205d158f722a495c6b3bbd77439864c075ea5db2d42331d1dcc                                   0.0s
=> => extracting sha256:8f7ead5f2dc0fb85f6d6b3c81284ccd4b80b783030d202a18ba5ccc5ee98f451                                   0.0s
=> => extracting sha256:b16687ec2b7bf8ddce81f21d836d6f9821f1b864f3e8e7fad1734d8295cb4a12                                   0.0s
=> => extracting sha256:f79056b7935ee1a11822640a4e32333a6fccfc86efddab21fd6d7300edf2d054                                   0.0s
=> => extracting sha256:400c1de759f415b6b2a0ba96bcef3e516f26171d5930757dcddc3d6971bf3e81                                   0.5s
=> [frontend internal] load build context                                                                                  0.2s
=> => transferring context: 74.89kB                                                                                        0.1s
=> [backend internal] load build context                                                                                   0.4s
=> => transferring context: 14.56kB                                                                                        0.3s
=> [backend builder 1/4] FROM docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc1  15.8s
=> => resolve docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11b1   0.1s
=> => sha256:bbc5a12237474ec0b23b5051260d56d6436a24d70d9cad961adebe7a037ed813 249B / 249B                                  0.3s
=> => sha256:6912e23eb44595b44dc6d9882b34f305ad6f6e9667cec70b13e5dbc4f8cd97f4 12.12MB / 12.12MB                            5.2s
=> => sha256:75b6a36c64a1cf3520fc4b4aae4739d5caa8a6a05c27835a920a61784c9c32f8 1.29MB / 1.29MB                              2.9s
=> => sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e 29.84MB / 29.84MB                           12.8s
=> => extracting sha256:ecc510c1e359bdc007b39802570bb7e4dec7d6ebccf357c3201a74299113e05e                                   1.2s
=> => extracting sha256:75b6a36c64a1cf3520fc4b4aae4739d5caa8a6a05c27835a920a61784c9c32f8                                   0.1s
=> => extracting sha256:6912e23eb44595b44dc6d9882b34f305ad6f6e9667cec70b13e5dbc4f8cd97f4                                   0.6s
=> => extracting sha256:bbc5a12237474ec0b23b5051260d56d6436a24d70d9cad961adebe7a037ed813                                   0.0s
=> [frontend stage-1 2/4] RUN apk upgrade --no-cache                                                                       7.3s
=> [backend builder 2/4] WORKDIR /build                                                                                    0.6s
```

Evidence group: Docker and Compose | File: m4-compose-build-2(2).png

## A07. Docker Compose build evidence 3



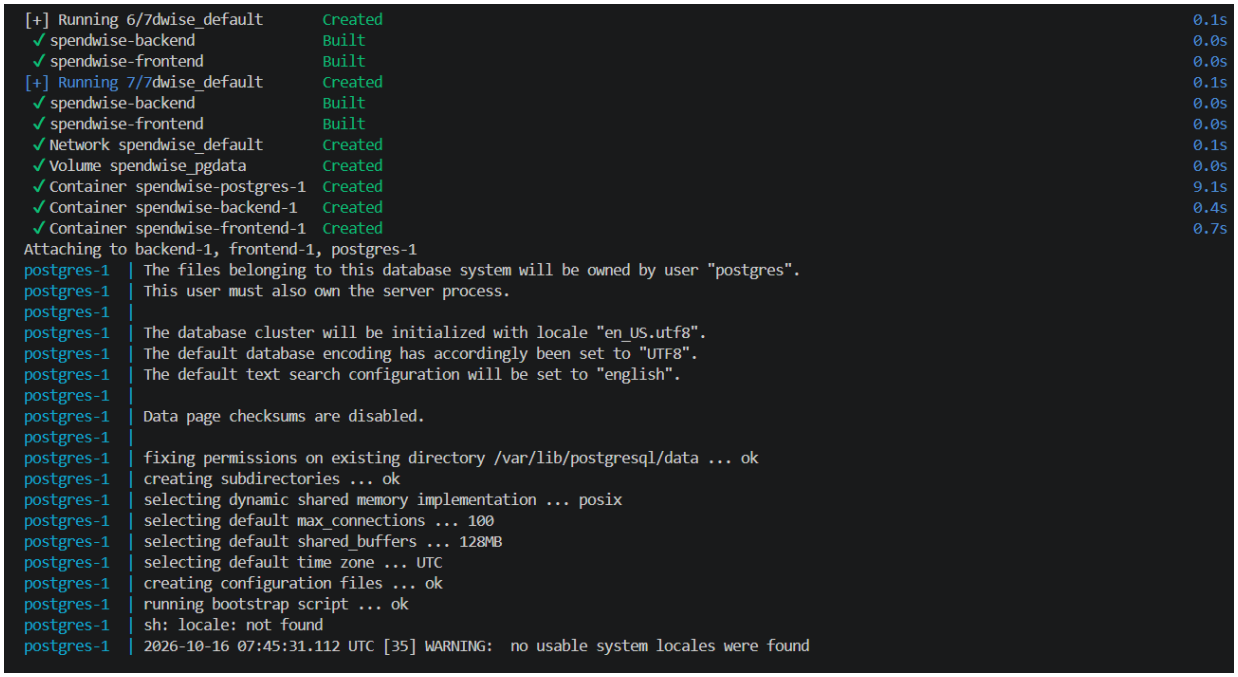

Evidence group: Docker and Compose | File: m4-compose-build-3(2).png

## A08. Docker Compose build evidence 4

```
 => => exporting attestation manifest sha256:67ea30372b175a037166b591f7e035646e6e88c216e65c075c21780d60fb04da                    0.1s
 => => exporting manifest list sha256:b1ef20f1a0946afebb65892f7d5ede475c87f869560feb7f87bb4487deb8f6f4                           0.1s
 => => naming to docker.io/library/spendwise-backend:latest                                                                      0.0s
 => => unpacking to docker.io/library/spendwise-backend:latest                                                                   2.2s
 => [backend] resolving provenance for metadata file                                                                             0.0s
[+] Running 4/5
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
[+] Running 4/5ckend              Built                                                                                         0.0s
```

Evidence group: Docker and Compose | File: m4-compose-build-4(2).png

## A09. Containers created by Docker Compose



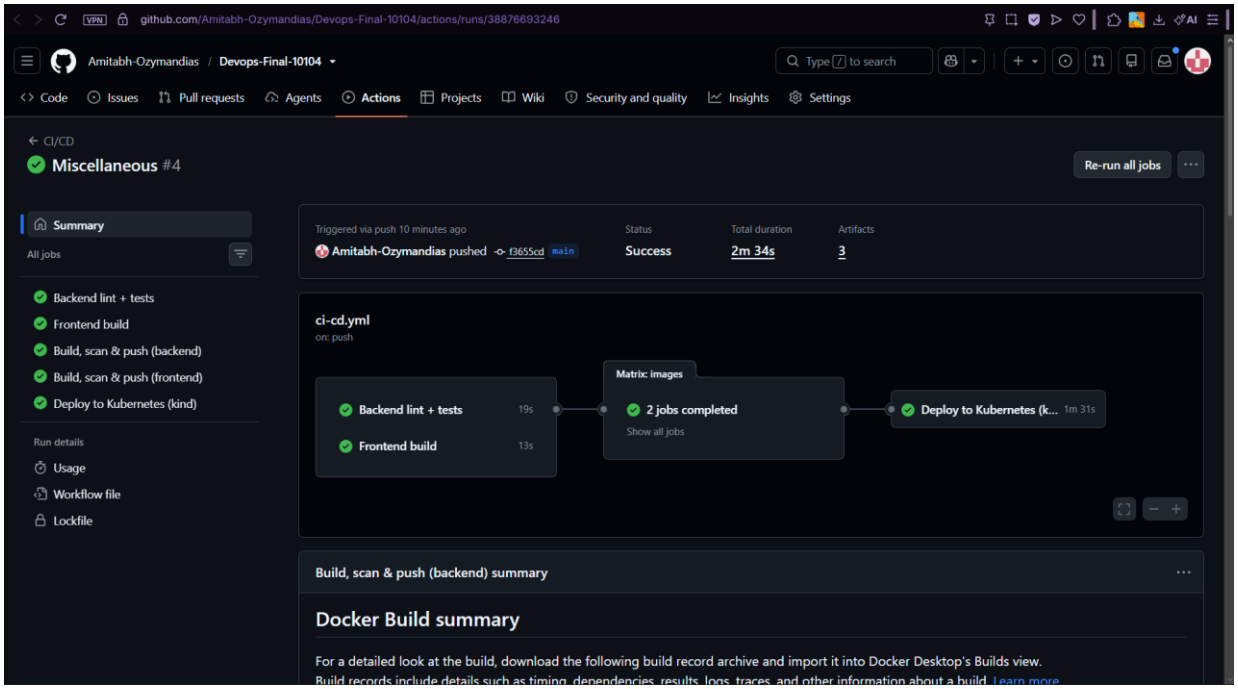

Evidence group: Docker and Compose | File: m4-compose-containers-created(2).png

## A10. Frontend container logs

```
postgres-1  | 2026-10-16 07:45:39.534 UTC [1] LOG:  listening on IPv6 address "::", port 5432
postgres-1  | 2026-10-16 07:45:39.565 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
postgres-1  | 2026-10-16 07:45:39.621 UTC [63] LOG:  database system was shut down at 2026-10-16 07:45:39 UTC
postgres-1  | 2026-10-16 07:45:39.652 UTC [1] LOG:  database system is ready to accept connections
frontend-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
frontend-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontend-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
frontend-1  | 10-listen-on-ipv6-by-default.sh: info: /etc/nginx/conf.d/default.conf differs from the packaged version
frontend-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
frontend-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
frontend-1  | 20-envsubst-on-templates.sh: Running envsubst on /etc/nginx/templates/default.conf.template to /etc/nginx/conf.d/default.conf
frontend-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
frontend-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: using the "epoll" event method
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: nginx/1.27.5
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: built by gcc 14.2.0 (Alpine 14.2.0)
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker processes
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 35
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 36
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 37
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 38
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 39
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 40
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 41
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 42
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 43
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 44
frontend-1  | 2026/10/16 07:45:41 [notice] 1#1: start worker process 45
```

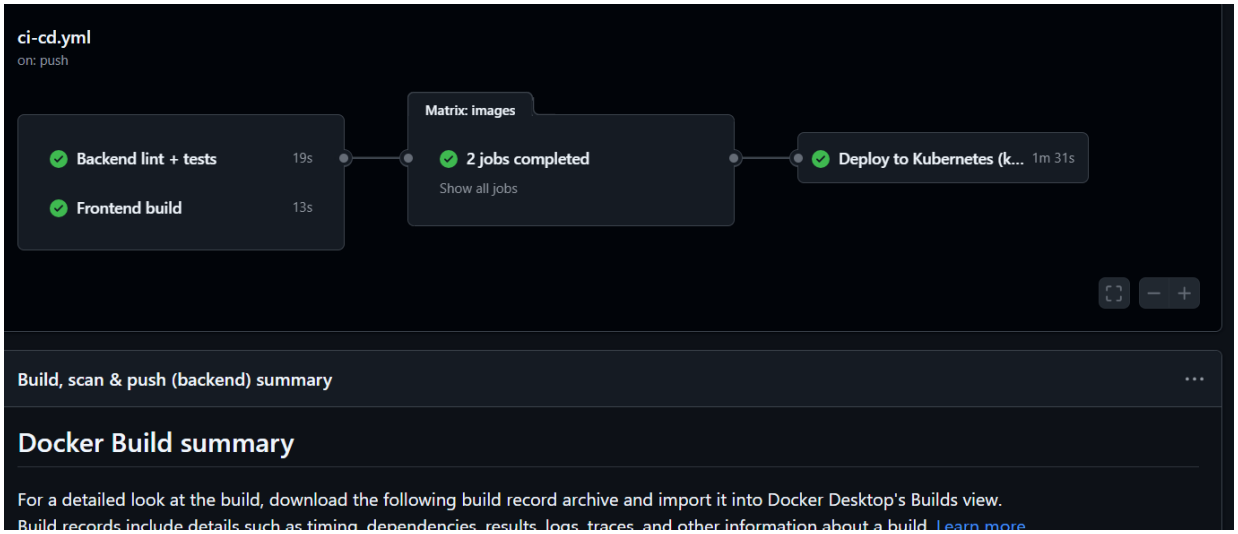
Evidence group: Docker and Compose | File: m4-compose-frontend-logs(2).png

## A11. Successful GitHub Actions workflow run



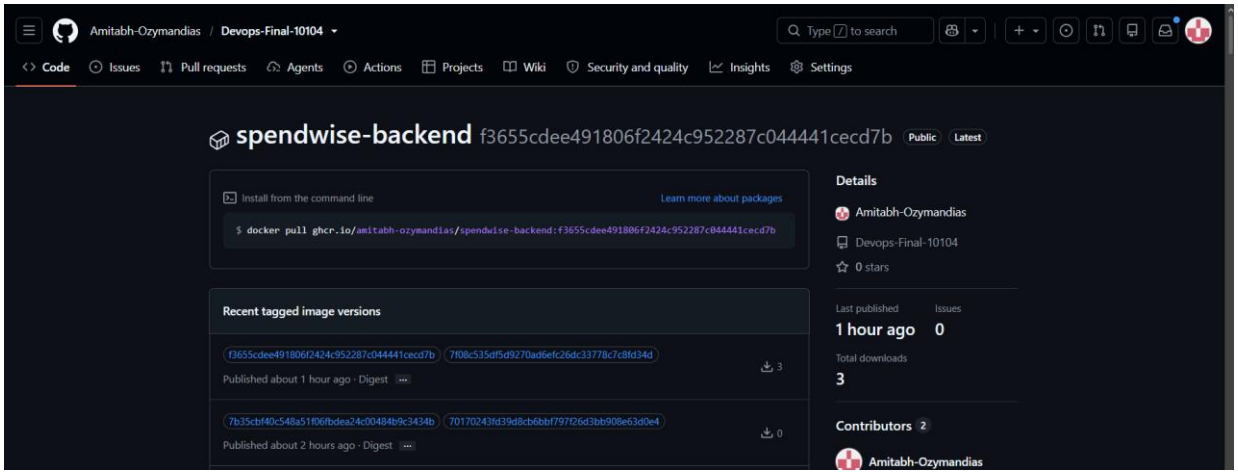

Evidence group: CI/CD and Image Provenance | File: m5-pipeline-run-green(2).png

## A12. CI/CD pipeline job graph



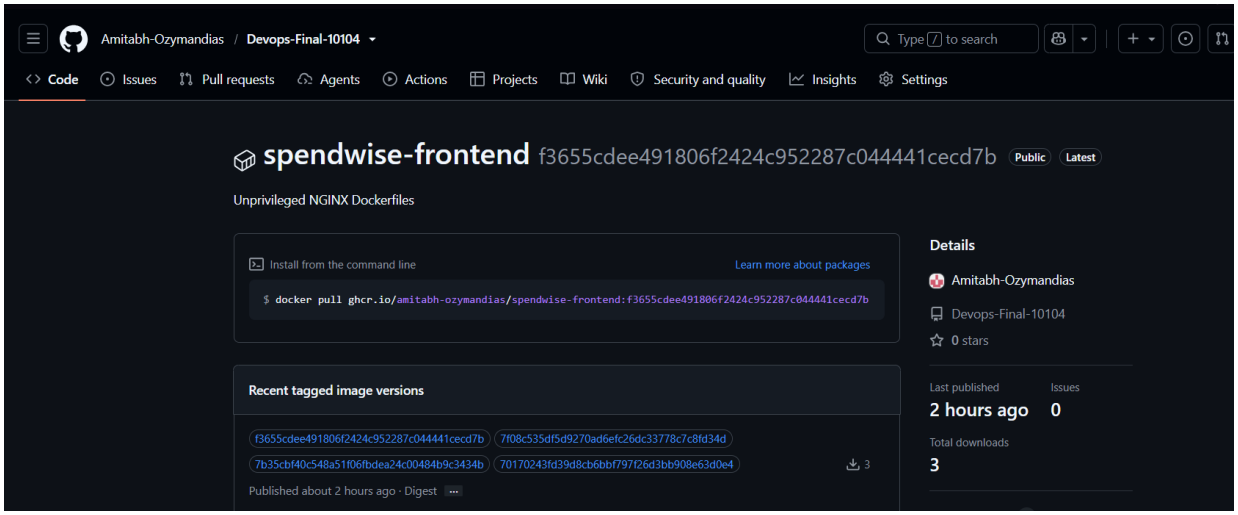

Evidence group: CI/CD and Image Provenance | File: m5-pipeline-graph(2).png

## A13. Backend images tagged by commit SHA in GHCR



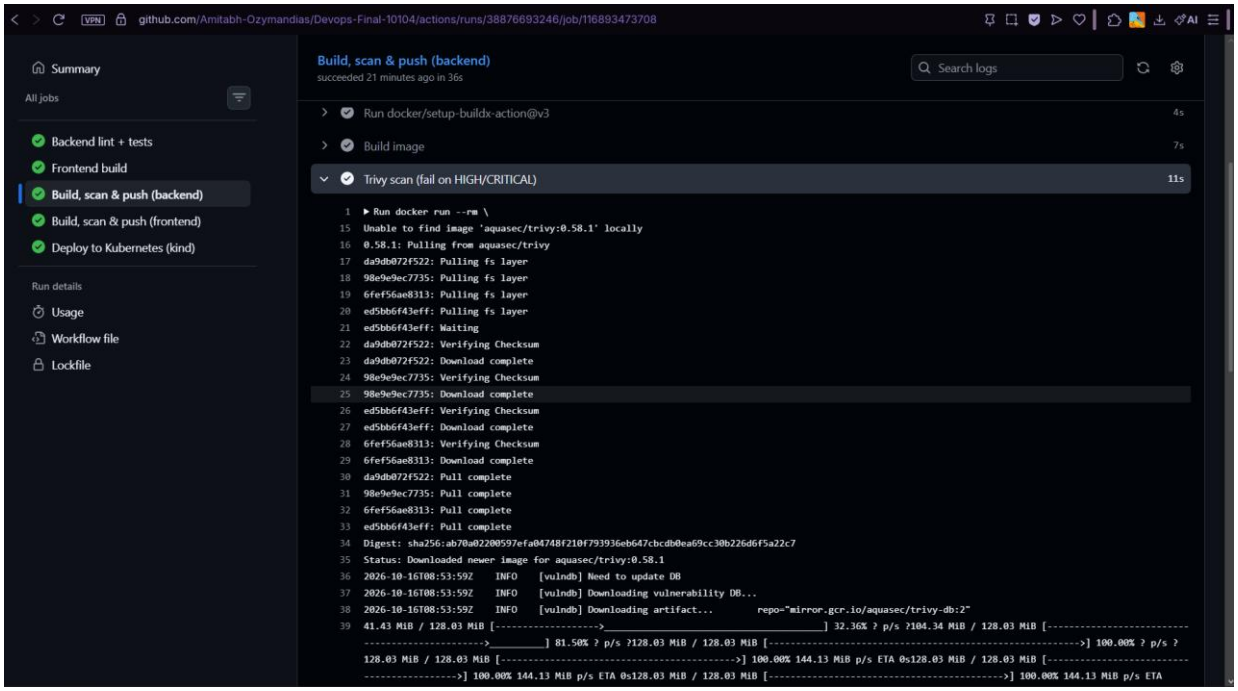

Evidence group: CI/CD and Image Provenance | File: m5-ghcr-backend-sha-tags(2).png

## A14. Frontend images tagged by commit SHA in GHCR



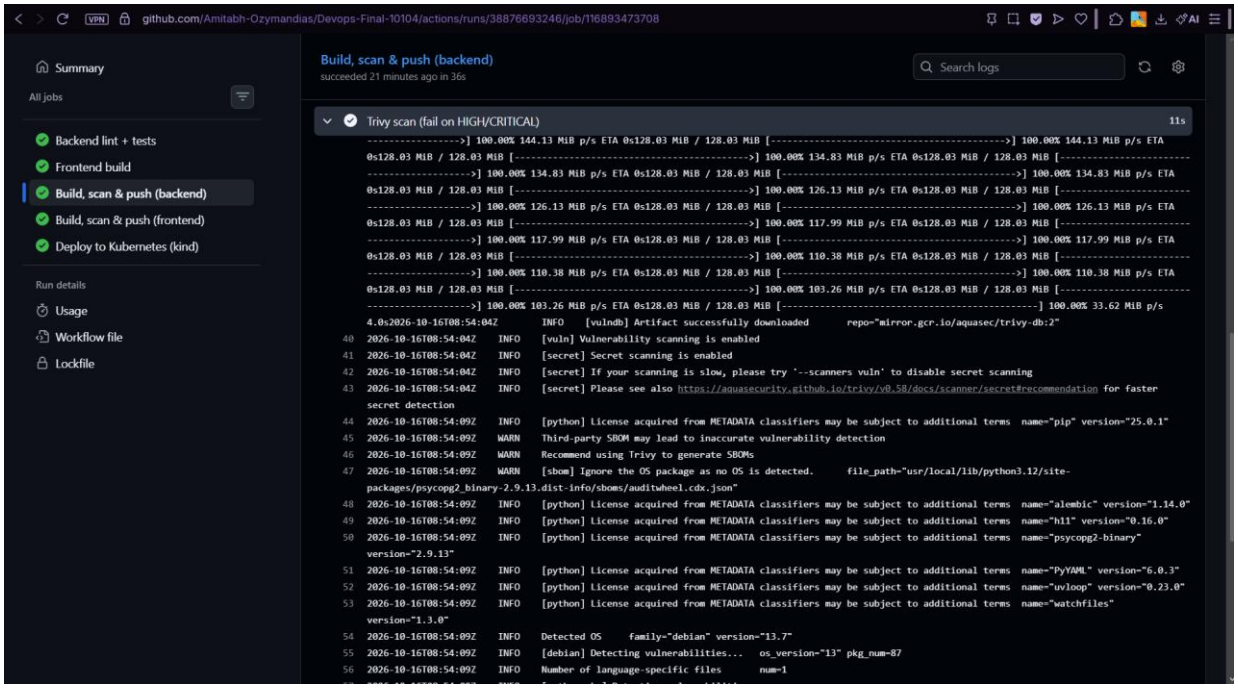

Evidence group: CI/CD and Image Provenance | File: m5-ghcr-frontend-sha-tags(2).png

## A15. Trivy backend image scan start



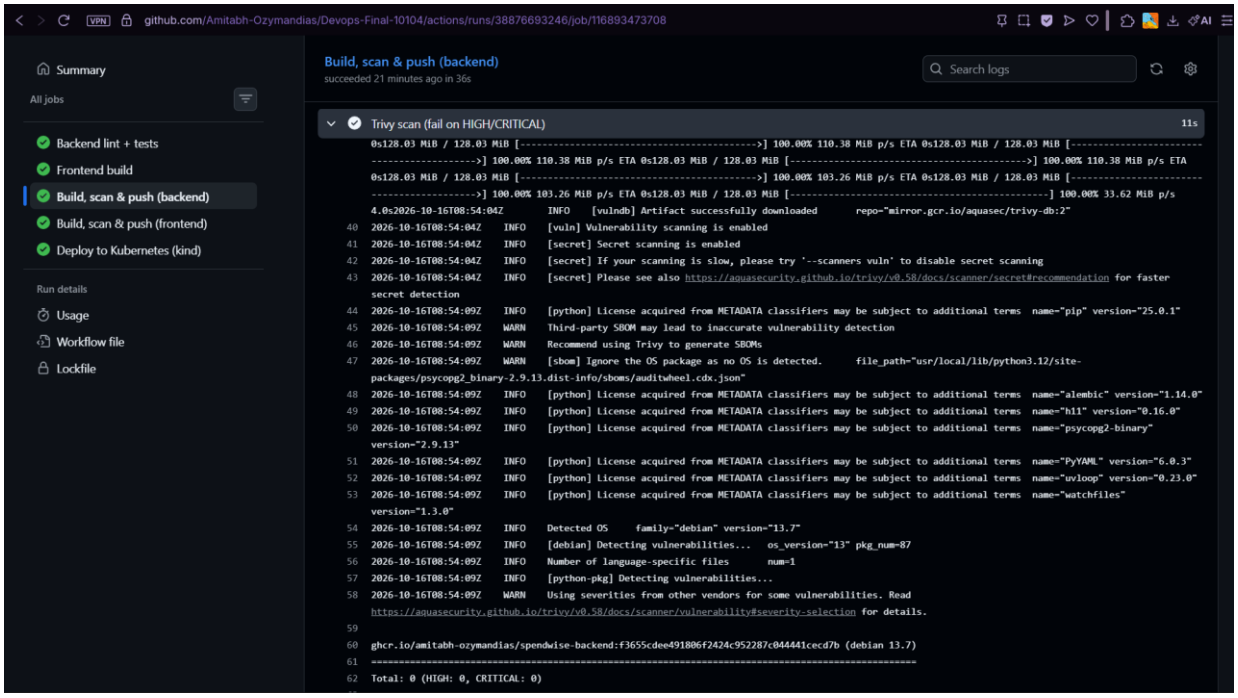

Evidence group: DevSecOps | File: m6-trivy-backend-1-start.png

## A16. Trivy backend image scanning output




Evidence group: DevSecOps | File: m6-trivy-backend-2-scanning.png

## A17. Trivy scan result after remediation




Evidence group: DevSecOps | File: m6-trivy-backend-3-result.png

## A18. Terraform validation and plan

```
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terraform$ cd "/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terra
form"
sed -i 's/^aws_region.*/aws_region = "ap-southeast-2"/' terraform.tfvars
terraform fmt                    # tidies the spacing in terraform.tfvars
grep -E "aws_region|create_eks" terraform.tfvars   # aws_region = "ap-southeast-2", create_eks = false
terraform validate               # Success! The configuration is valid.
terraform plan                   # ... should end: "Plan: 9 to add, 0 to change, 0 to destroy"
terraform.tfvars
aws_region   = "ap-southeast-2"
create_eks          = false
Success! The configuration is valid.

data.aws_availability_zones.available: Reading...
data.aws_availability_zones.available: Read complete after 2s [id=ap-southeast-2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # module.vpc.aws_default_network_acl.this[0] will be created
  + resource "aws_default_network_acl" "this" {
      + arn                    = (known after apply)
      + default_network_acl_id = (known after apply)
      + id                     = (known after apply)
      + owner_id               = (known after apply)
      + tags                   = {
          + "Name" = "spendwise-vpc-default"
        }
      + tags_all               = {
          + "Environment" = "dev"
          + "ManagedBy"   = "terraform"
          + "Name"        = "spendwise-vpc-default"
          + "Project"     = "spendwise"
        }
      + vpc_id                 = (known after apply)
```

Evidence group: Terraform | File: m7-terraform-plan-1-validate.png

## A19. Terraform plan evidence 2

```
      + egress {
          + action          = "allow"
          + from_port       = 0
          + ipv6_cidr_block = "::/0"
          + protocol        = "-1"
          + rule_no         = 101
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }
      + egress {
          + action          = "allow"
          + cidr_block      = "0.0.0.0/0"
          + from_port       = 0
          + protocol        = "-1"
          + rule_no         = 100
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }

      + ingress {
          + action          = "allow"
          + from_port       = 0
          + ipv6_cidr_block = "::/0"
          + protocol        = "-1"
          + rule_no         = 101
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }
      + ingress {
          + action          = "allow"
          + cidr_block      = "0.0.0.0/0"
          + from_port       = 0
          + protocol        = "-1"
          + rule_no         = 100
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }
```

Evidence group: Terraform | File: m7-terraform-plan-2.png

## A20. Terraform plan evidence 3

```
# module.vpc.aws_default_route_table.default[0] will be created
+ resource "aws_default_route_table" "default" {
    + arn                    = (known after apply)
    + default_route_table_id = (known after apply)
    + id                     = (known after apply)
    + owner_id               = (known after apply)
    + route                  = (known after apply)
    + tags                   = {
        + "Name" = "spendwise-vpc-default"
      }
    + tags_all               = {
        + "Environment" = "dev"
        + "ManagedBy"   = "terraform"
        + "Name"        = "spendwise-vpc-default"
        + "Project"     = "spendwise"
      }
    + vpc_id                 = (known after apply)

    + timeouts {
        + create = "5m"
        + update = "5m"
      }
  }

# module.vpc.aws_default_security_group.this[0] will be created
+ resource "aws_default_security_group" "this" {
    + arn                    = (known after apply)
    + description            = (known after apply)
    + egress                 = (known after apply)
    + id                     = (known after apply)
    + ingress                = (known after apply)
    + name                   = (known after apply)
    + name_prefix            = (known after apply)
    + owner_id               = (known after apply)
    + revoke_rules_on_delete = false
    + tags                   = {
```

Evidence group: Terraform | File: m7-terraform-plan-3.png

## A21. Terraform plan evidence 4

```
      + "Name" = "spendwise-vpc-default"
    }
  + tags_all               = {
      + "Environment" = "dev"
      + "ManagedBy"   = "terraform"
      + "Name"        = "spendwise-vpc-default"
      + "Project"     = "spendwise"
    }
  + vpc_id                 = (known after apply)
}

# module.vpc.aws_internet_gateway.this[0] will be created
+ resource "aws_internet_gateway" "this" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + tags     = {
      + "Name" = "spendwise-vpc"
    }
  + tags_all = {
      + "Environment" = "dev"
      + "ManagedBy"   = "terraform"
      + "Name"        = "spendwise-vpc"
      + "Project"     = "spendwise"
    }
  + vpc_id   = (known after apply)
}

# module.vpc.aws_route.public_internet_gateway[0] will be created
+ resource "aws_route" "public_internet_gateway" {
  + destination_cidr_block = "0.0.0.0/0"
  + gateway_id             = (known after apply)
  + id                     = (known after apply)
  + instance_id            = (known after apply)
  + instance_owner_id      = (known after apply)
  + network_interface_id   = (known after apply)
  + origin                 = (known after apply)
```

Evidence group: Terraform | File: m7-terraform-plan-4.png

## A22. Terraform plan evidence 5

```
      + route_table_id        = (known after apply)
      + state                 = (known after apply)

      + timeouts {
          + create = "5m"
        }
    }

  # module.vpc.aws_route_table.public[0] will be created
  + resource "aws_route_table" "public" {
      + arn              = (known after apply)
      + id               = (known after apply)
      + owner_id         = (known after apply)
      + propagating_vgws = (known after apply)
      + route            = (known after apply)
      + tags             = {
          + "Name" = "spendwise-vpc-public"
        }
      + tags_all         = {
          + "Environment" = "dev"
          + "ManagedBy"   = "terraform"
          + "Name"        = "spendwise-vpc-public"
          + "Project"     = "spendwise"
        }
      + vpc_id           = (known after apply)
    }

  # module.vpc.aws_route_table_association.public[0] will be created
  + resource "aws_route_table_association" "public" {
      + id             = (known after apply)
      + route_table_id = (known after apply)
      + subnet_id      = (known after apply)
    }

  # module.vpc.aws_route_table_association.public[1] will be created
  + resource "aws_route_table_association" "public" {
      + id             = (known after apply)
```

Evidence group: Terraform | File: m7-terraform-plan-5.png

## A23. Subnet A configuration and plan

```
      + route_table_id = (known after apply)
      + subnet_id      = (known after apply)
    }

  # module.vpc.aws_subnet.public[0] will be created
  + resource "aws_subnet" "public" {
      + arn                                            = (known after apply)
      + assign_ipv6_address_on_creation                = false
      + availability_zone                              = "ap-southeast-2a"
      + availability_zone_id                           = (known after apply)
      + cidr_block                                     = "10.20.0.0/24"
      + enable_dns64                                   = false
      + enable_resource_name_dns_a_record_on_launch    = false
      + enable_resource_name_dns_aaaa_record_on_launch = false
      + id                                             = (known after apply)
      + ipv6_cidr_block_association_id                 = (known after apply)
      + ipv6_native                                    = false
      + map_public_ip_on_launch                        = true
      + owner_id                                       = (known after apply)
      + private_dns_hostname_type_on_launch            = (known after apply)
      + tags                                           = {
          + "Name"                                = "spendwise-vpc-public-ap-southeast-2a"
          + "kubernetes.io/cluster/spendwise-eks" = "shared"
          + "kubernetes.io/role/elb"              = "1"
        }
      + tags_all                                       = {
          + "Environment"                         = "dev"
          + "ManagedBy"                           = "terraform"
          + "Name"                                = "spendwise-vpc-public-ap-southeast-2a"
          + "Project"                             = "spendwise"
          + "kubernetes.io/cluster/spendwise-eks" = "shared"
          + "kubernetes.io/role/elb"              = "1"
        }
      + vpc_id                                         = (known after apply)
    }

  # module.vpc.aws_subnet.public[1] will be created
```

Evidence group: Terraform | File: m7-terraform-plan-6-subnet-a.png

## A24. Subnet B configuration and plan

```
  + resource "aws_subnet" "public" {
      + arn                                            = (known after apply)
      + assign_ipv6_address_on_creation                = false
      + availability_zone                              = "ap-southeast-2b"
      + availability_zone_id                           = (known after apply)
      + cidr_block                                     = "10.20.1.0/24"
      + enable_dns64                                   = false
      + enable_resource_name_dns_a_record_on_launch    = false
      + enable_resource_name_dns_aaaa_record_on_launch = false
      + id                                             = (known after apply)
      + ipv6_cidr_block_association_id                 = (known after apply)
      + ipv6_native                                    = false
      + map_public_ip_on_launch                        = true
      + owner_id                                       = (known after apply)
      + private_dns_hostname_type_on_launch            = (known after apply)
      + tags                                           = {
          + "Name"                                = "spendwise-vpc-public-ap-southeast-2b"
          + "kubernetes.io/cluster/spendwise-eks" = "shared"
          + "kubernetes.io/role/elb"              = "1"
        }
      + tags_all                                       = {
          + "Environment"                         = "dev"
          + "ManagedBy"                           = "terraform"
          + "Name"                                = "spendwise-vpc-public-ap-southeast-2b"
          + "Project"                             = "spendwise"
          + "kubernetes.io/cluster/spendwise-eks" = "shared"
          + "kubernetes.io/role/elb"              = "1"
        }
      + vpc_id                                         = (known after apply)
    }

  # module.vpc.aws_vpc.this[0] will be created
  + resource "aws_vpc" "this" {
      + arn                                = (known after apply)
      + cidr_block                         = "10.20.0.0/16"
      + default_network_acl_id             = (known after apply)
      + default_route_table_id             = (known after apply)
```

Evidence group: Terraform | File: m7-terraform-plan-7-subnet-b.png

## A25. Terraform plan summary

```
      + cidr_block                           = "10.20.0.0/16"
      + default_network_acl_id               = (known after apply)
      + default_route_table_id               = (known after apply)
      + default_security_group_id            = (known after apply)
      + dhcp_options_id                      = (known after apply)
      + enable_dns_hostnames                 = true
      + enable_dns_support                   = true
      + enable_network_address_usage_metrics = (known after apply)
      + id                                   = (known after apply)
      + instance_tenancy                     = "default"
      + ipv6_association_id                  = (known after apply)
      + ipv6_cidr_block                      = (known after apply)
      + ipv6_cidr_block_network_border_group = (known after apply)
      + main_route_table_id                  = (known after apply)
      + owner_id                             = (known after apply)
      + tags                                 = {
          + "Name" = "spendwise-vpc"
        }
      + tags_all                             = {
          + "Environment" = "dev"
          + "ManagedBy"   = "terraform"
          + "Name"        = "spendwise-vpc"
          + "Project"     = "spendwise"
        }
    }

Plan: 11 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + configure_kubectl = "EKS not created (create_eks = false)"
  + public_subnet_ids = [
      + (known after apply),
      + (known after apply),
    ]
  + region            = "ap-southeast-2"
  + vpc_id            = (known after apply)
```

Evidence group: Terraform | File: m7-terraform-plan-8-summary.png

## A26. Terraform apply started

```
Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform
apply" now.
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terraform$ terraform apply
data.aws_availability_zones.available: Reading...
data.aws_availability_zones.available: Read complete after 5s [id=ap-southeast-2]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # module.vpc.aws_default_network_acl.this[0] will be created
  + resource "aws_default_network_acl" "this" {
      + arn                    = (known after apply)
      + default_network_acl_id = (known after apply)
      + id                     = (known after apply)
      + owner_id               = (known after apply)
      + tags                   = {
          + "Name" = "spendwise-vpc-default"
        }
      + tags_all               = {
          + "Environment" = "dev"
          + "ManagedBy"   = "terraform"
          + "Name"        = "spendwise-vpc-default"
          + "Project"     = "spendwise"
        }
      + vpc_id                 = (known after apply)

      + egress {
          + action          = "allow"
          + from_port       = 0
          + ipv6_cidr_block = "::/0"
          + protocol        = "-1"
          + rule_no         = 101
          + to_port         = 0
            # (1 unchanged attribute hidden)
        }
```

Evidence group: Terraform | File: m7-terraform-apply-1-start.png

## A27. Terraform resources being created

```
Plan: 11 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + configure_kubectl = "EKS not created (create_eks = false)"
  + public_subnet_ids = [
      + (known after apply),
      + (known after apply),
    ]
  + region            = "ap-southeast-2"
  + vpc_id            = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

module.vpc.aws_vpc.this[0]: Creating...
module.vpc.aws_vpc.this[0]: Still creating... [00m10s elapsed]
module.vpc.aws_vpc.this[0]: Still creating... [00m20s elapsed]
module.vpc.aws_vpc.this[0]: Creation complete after 21s [id=vpc-0927ec589e5d6629b]
module.vpc.aws_internet_gateway.this[0]: Creating...
module.vpc.aws_subnet.public[1]: Creating...
module.vpc.aws_subnet.public[0]: Creating...
module.vpc.aws_route_table.public[0]: Creating...
module.vpc.aws_default_route_table.default[0]: Creating...
module.vpc.aws_default_security_group.this[0]: Creating...
module.vpc.aws_default_network_acl.this[0]: Creating...
module.vpc.aws_default_route_table.default[0]: Creation complete after 5s [id=rtb-0a51f230293a079f0]
module.vpc.aws_internet_gateway.this[0]: Creation complete after 5s [id=igw-057db52e7242e210a]
module.vpc.aws_route_table.public[0]: Creation complete after 5s [id=rtb-0c22c73d63fc93403]
module.vpc.aws_route.public_internet_gateway[0]: Creating...
module.vpc.aws_route.public_internet_gateway[0]: Creation complete after 2s [id=r-rtb-0c22c73d63fc934031080289494]
module.vpc.aws_default_security_group.this[0]: Creation complete after 7s [id=sg-0df58690ddab809c6]
module.vpc.aws_default_network_acl.this[0]: Creation complete after 8s [id=acl-059b6521cb992d59a]
module.vpc.aws_subnet.public[1]: Still creating... [00m10s elapsed]
module.vpc.aws_subnet.public[0]: Still creating... [00m10s elapsed]
```

Evidence group: Terraform | File: m7-terraform-apply-2-creating.png

## A28. Terraform apply completed

```
module.vpc.aws_subnet.public[1]: Creation complete after 16s [id=subnet-08465a1f681a4f0db]
module.vpc.aws_subnet.public[0]: Creation complete after 16s [id=subnet-04672d1cf8b7786cb]
module.vpc.aws_route_table_association.public[0]: Creating...
module.vpc.aws_route_table_association.public[1]: Creating...
module.vpc.aws_route_table_association.public[1]: Creation complete after 1s [id=rtbassoc-06ab6f2e453e0cbe5]
module.vpc.aws_route_table_association.public[0]: Creation complete after 1s [id=rtbassoc-05fd40341b66731a4]

Apply complete! Resources: 11 added, 0 changed, 0 destroyed.

Outputs:

configure_kubectl = "EKS not created (create_eks = false)"
public_subnet_ids = [
  "subnet-04672d1cf8b7786cb",
  "subnet-08465a1f681a4f0db",
]
region = "ap-southeast-2"
vpc_id = "vpc-0927ec589e5d6629b"
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terraform$ terraform output        # 📸 shows vpc_id and the 2 public
_subnet_ids
configure_kubectl = "EKS not created (create_eks = false)"
public_subnet_ids = [
  "subnet-04672d1cf8b7786cb",
  "subnet-08465a1f681a4f0db",
]
region = "ap-southeast-2"
vpc_id = "vpc-0927ec589e5d6629b"
```

Evidence group: Terraform | File: m7-terraform-apply-3-complete.png

## A29. Terraform destroy started

```
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terraform$ terraform destroy
data.aws_availability_zones.available: Reading...
module.vpc.aws_vpc.this[0]: Refreshing state... [id=vpc-0927ec589e5d6629b]
data.aws_availability_zones.available: Read complete after 2s [id=ap-southeast-2]
module.vpc.aws_internet_gateway.this[0]: Refreshing state... [id=igw-057db52e7242e210a]
module.vpc.aws_subnet.public[0]: Refreshing state... [id=subnet-04672d1cf8b7786cb]
module.vpc.aws_subnet.public[1]: Refreshing state... [id=subnet-08465a1f681a4f0db]
module.vpc.aws_default_route_table.default[0]: Refreshing state... [id=rtb-0a51f230293a079f0]
module.vpc.aws_default_network_acl.this[0]: Refreshing state... [id=acl-059b6521cb992d59a]
module.vpc.aws_default_security_group.this[0]: Refreshing state... [id=sg-0df58690ddab809c6]
module.vpc.aws_route_table.public[0]: Refreshing state... [id=rtb-0c22c73d63fc93403]
module.vpc.aws_route_table_association.public[0]: Refreshing state... [id=rtbassoc-05fd40341b66731a4]
module.vpc.aws_route.public_internet_gateway[0]: Refreshing state... [id=r-rtb-0c22c73d63fc934031080289494]
module.vpc.aws_route_table_association.public[1]: Refreshing state... [id=rtbassoc-06ab6f2e453e0cbe5]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # module.vpc.aws_default_network_acl.this[0] will be destroyed
  - resource "aws_default_network_acl" "this" {
      - arn                    = "arn:aws:ec2:ap-southeast-2:929214127354:network-acl/acl-059b6521cb992d59a" -> null
      - default_network_acl_id = "acl-059b6521cb992d59a" -> null
      - id                     = "acl-059b6521cb992d59a" -> null
      - owner_id               = "929214127354" -> null
      - subnet_ids             = [
          - "subnet-04672d1cf8b7786cb",
          - "subnet-08465a1f681a4f0db",
        ] -> null
      - tags                   = {
          - "Name" = "spendwise-vpc-default"
        } -> null
      - tags_all               = {
```

Evidence group: Terraform | File: m7-terraform-destroy-1-start.png

## A30. Terraform destroy completed

```
  - region            = "ap-southeast-2" -> null
  - vpc_id            = "vpc-0927ec589e5d6629b" -> null

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

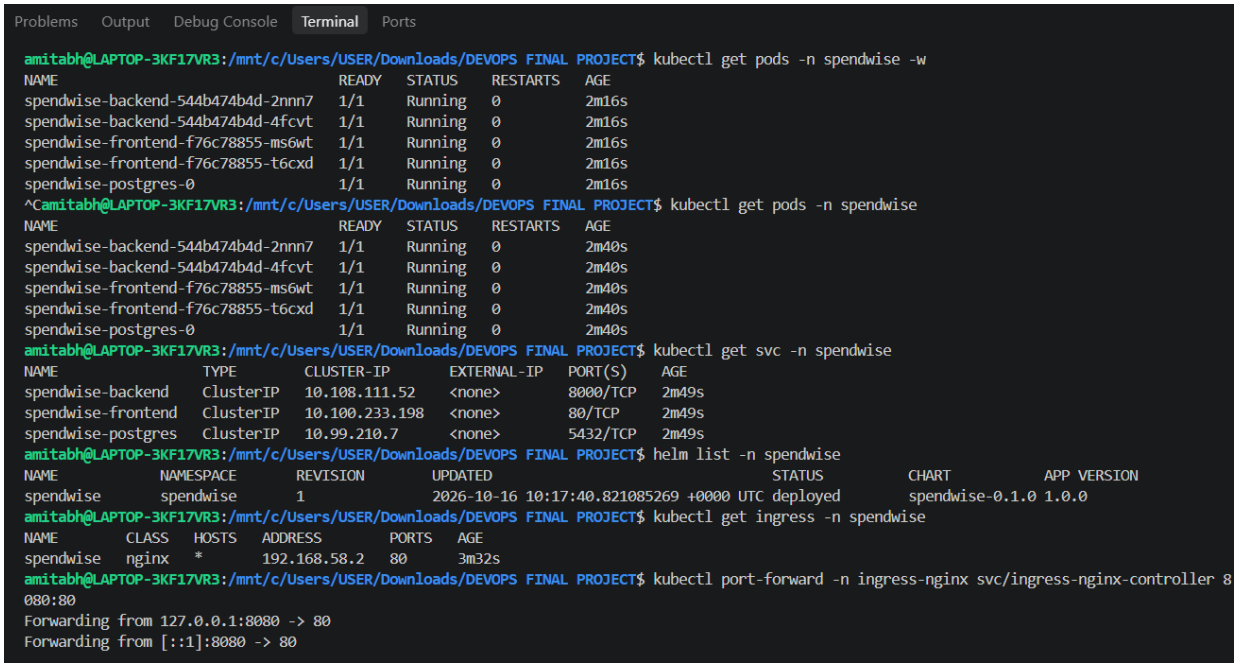
  Enter a value: yes

module.vpc.aws_default_network_acl.this[0]: Destroying... [id=acl-059b6521cb992d59a]
module.vpc.aws_default_security_group.this[0]: Destroying... [id=sg-0df58690ddab809c6]
module.vpc.aws_route_table_association.public[1]: Destroying... [id=rtbassoc-06ab6f2e453e0cbe5]
module.vpc.aws_route_table_association.public[0]: Destroying... [id=rtbassoc-05fd40341b66731a4]
module.vpc.aws_default_route_table.default[0]: Destroying... [id=rtb-0a51f230293a079f0]
module.vpc.aws_route.public_internet_gateway[0]: Destroying... [id=r-rtb-0c22c73d63fc934031080289494]
module.vpc.aws_default_route_table.default[0]: Destruction complete after 0s
module.vpc.aws_default_security_group.this[0]: Destruction complete after 0s
module.vpc.aws_default_network_acl.this[0]: Destruction complete after 0s
module.vpc.aws_route_table_association.public[1]: Destruction complete after 2s
module.vpc.aws_route_table_association.public[0]: Destruction complete after 2s
module.vpc.aws_subnet.public[1]: Destroying... [id=subnet-08465a1f681a4f0db]
module.vpc.aws_subnet.public[0]: Destroying... [id=subnet-04672d1cf8b7786cb]
module.vpc.aws_route.public_internet_gateway[0]: Destruction complete after 3s
module.vpc.aws_internet_gateway.this[0]: Destroying... [id=igw-057db52e7242e210a]
module.vpc.aws_route_table.public[0]: Destroying... [id=rtb-0c22c73d63fc93403]
module.vpc.aws_subnet.public[1]: Destruction complete after 2s
module.vpc.aws_subnet.public[0]: Destruction complete after 2s
module.vpc.aws_internet_gateway.this[0]: Destruction complete after 2s
module.vpc.aws_route_table.public[0]: Destruction complete after 3s
module.vpc.aws_vpc.this[0]: Destroying... [id=vpc-0927ec589e5d6629b]
module.vpc.aws_vpc.this[0]: Destruction complete after 1s

Destroy complete! Resources: 11 destroyed.
amitabh@LAPTOP-3KF17VR3:/mnt/c/Users/USER/Downloads/DEVOPS FINAL PROJECT/terraform$
```

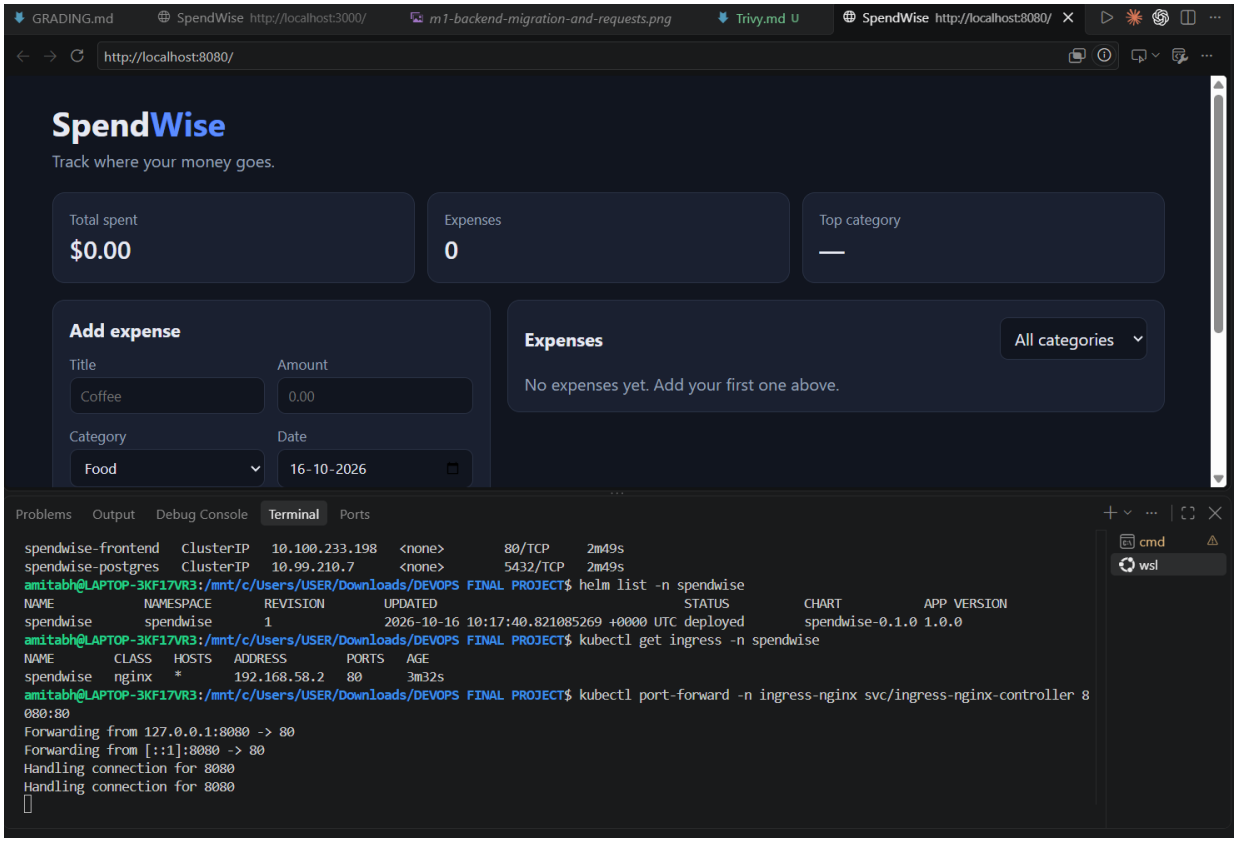
Evidence group: Terraform | File: m7-terraform-destroy-2-complete.png

## A31. Kubernetes pods, services, Helm and Ingress



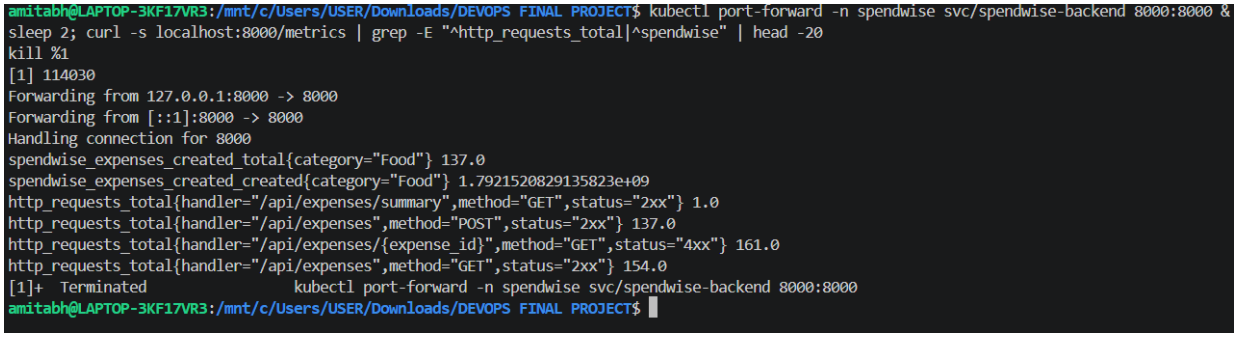

Evidence group: Kubernetes and Helm | File: m8-kubectl-pods-svc-helm-ingress.png

## A32. SpendWise application served through Ingress



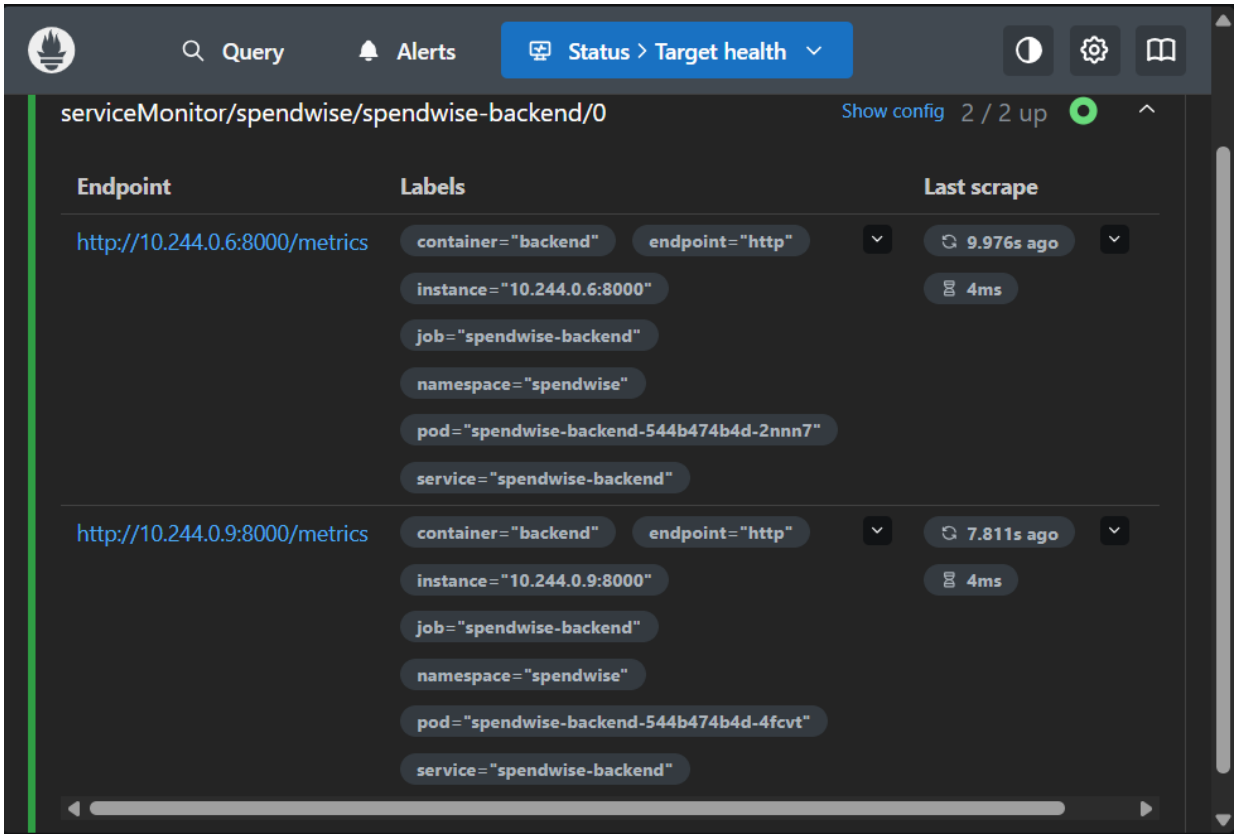

Evidence group: Kubernetes and Helm | File: m8-app-via-ingress.png

## A33. Backend Prometheus metrics endpoint



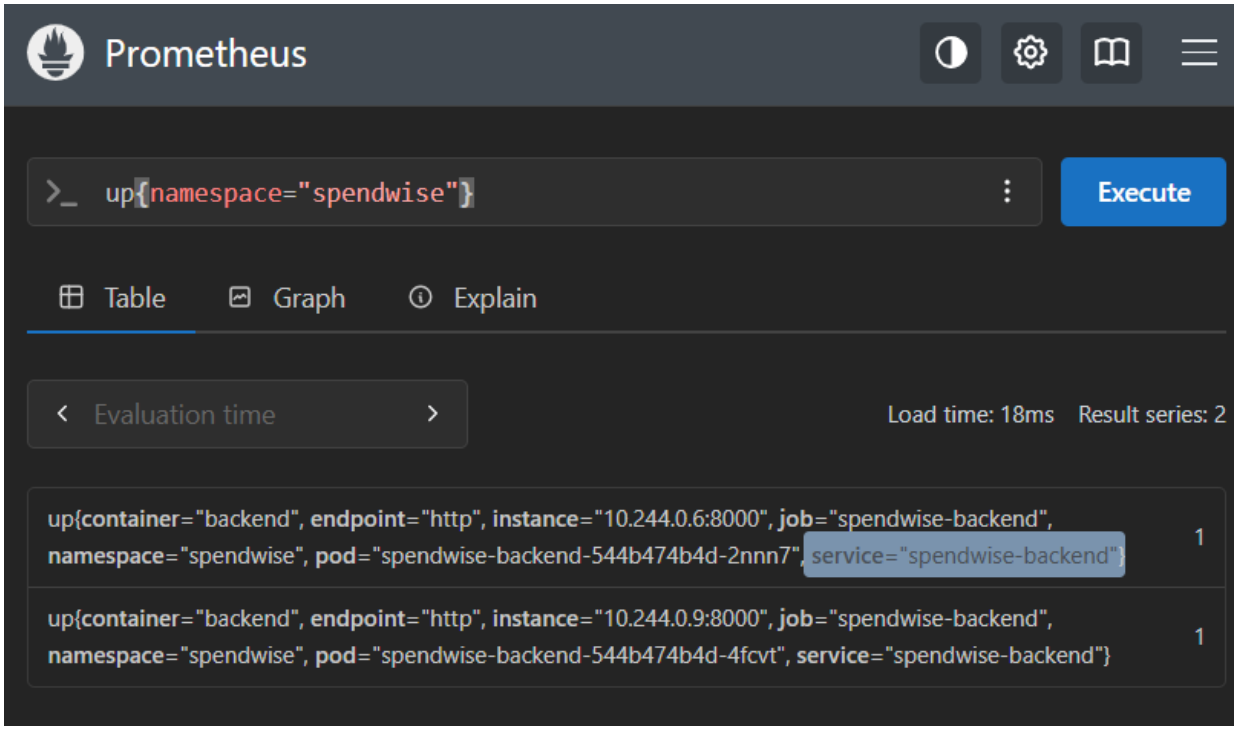

Evidence group: Observability | File: m9-metrics-endpoint.png

## A34. Prometheus targets for both SpendWise backend pods



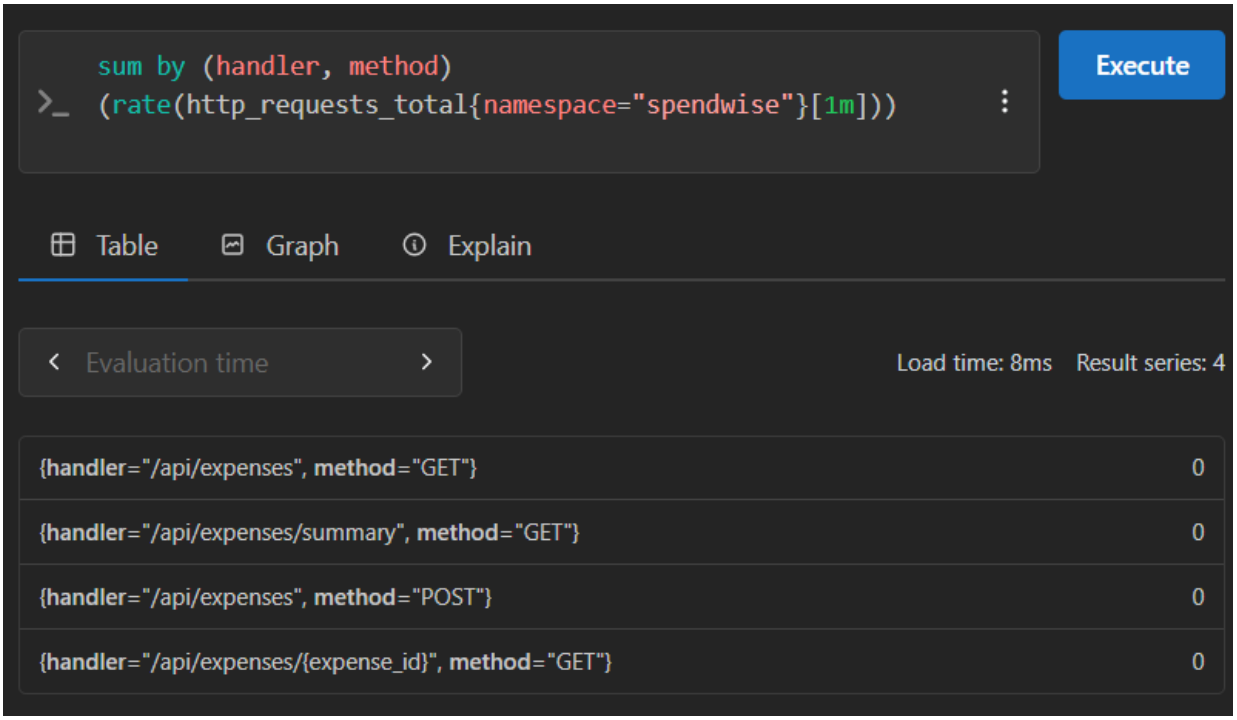

Evidence group: Observability | File: m9-prometheus-targets-spendwise.png

## A35. Prometheus up query for backend targets



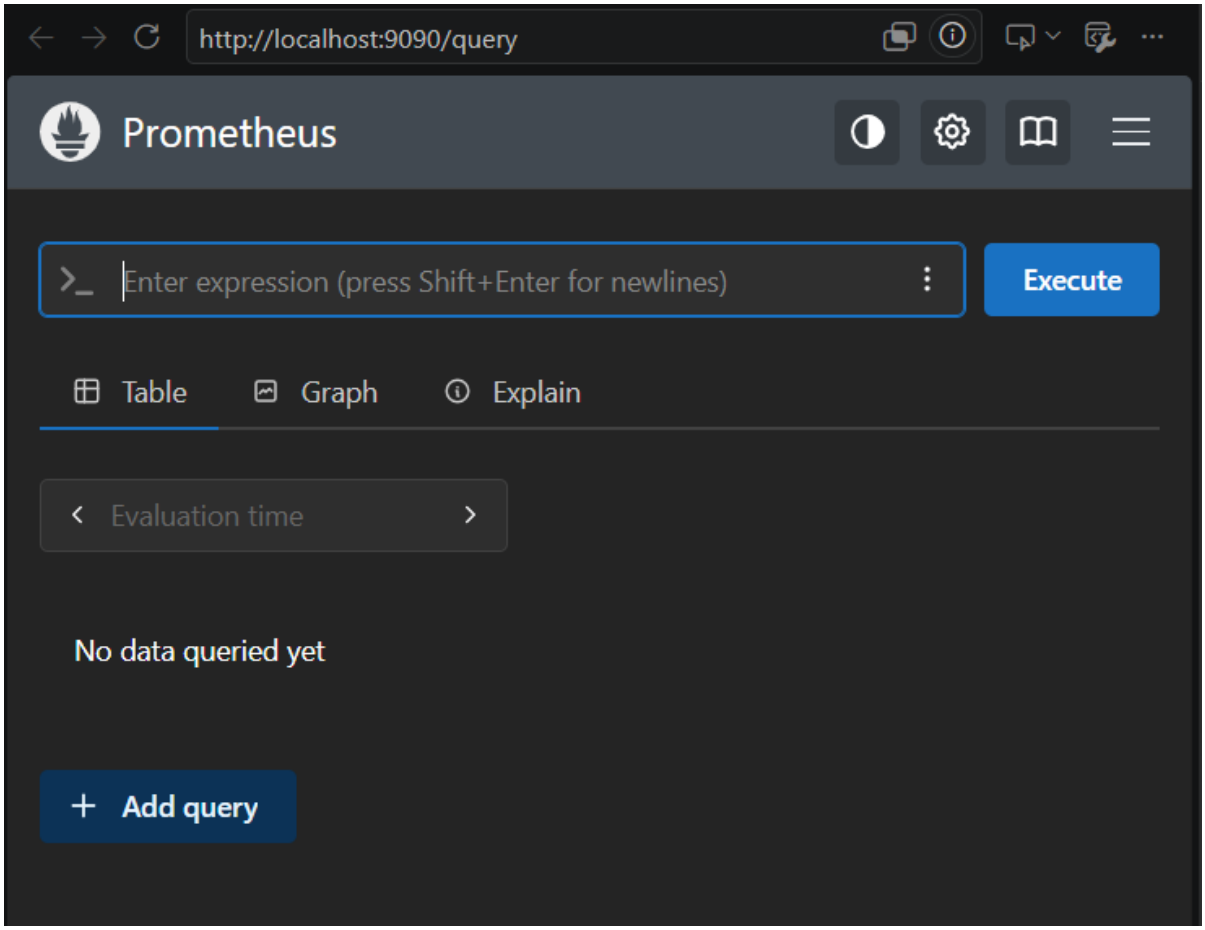

Evidence group: Observability | File: m9-prometheus-query-up(2).png

## A36. Prometheus request-rate query



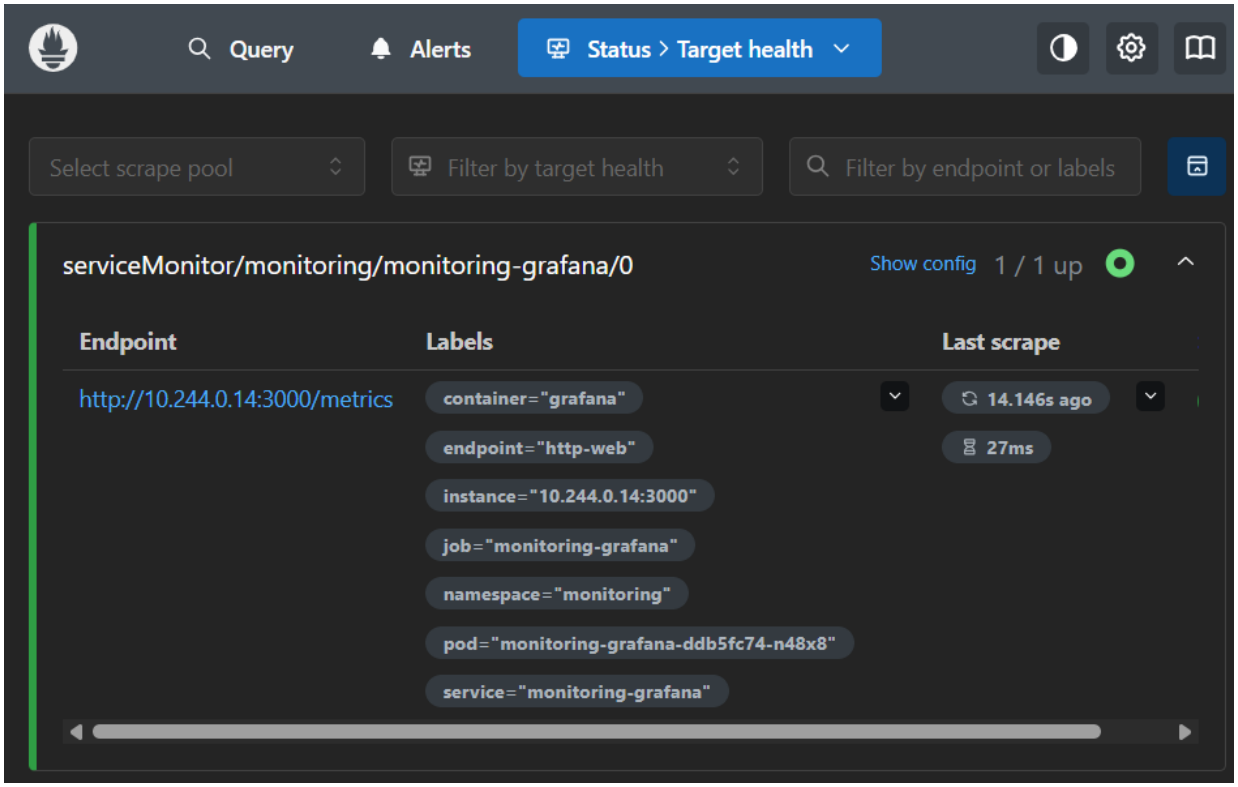

Evidence group: Observability | File: m9-prometheus-query-request-rate(2).png

## A37. Prometheus query interface



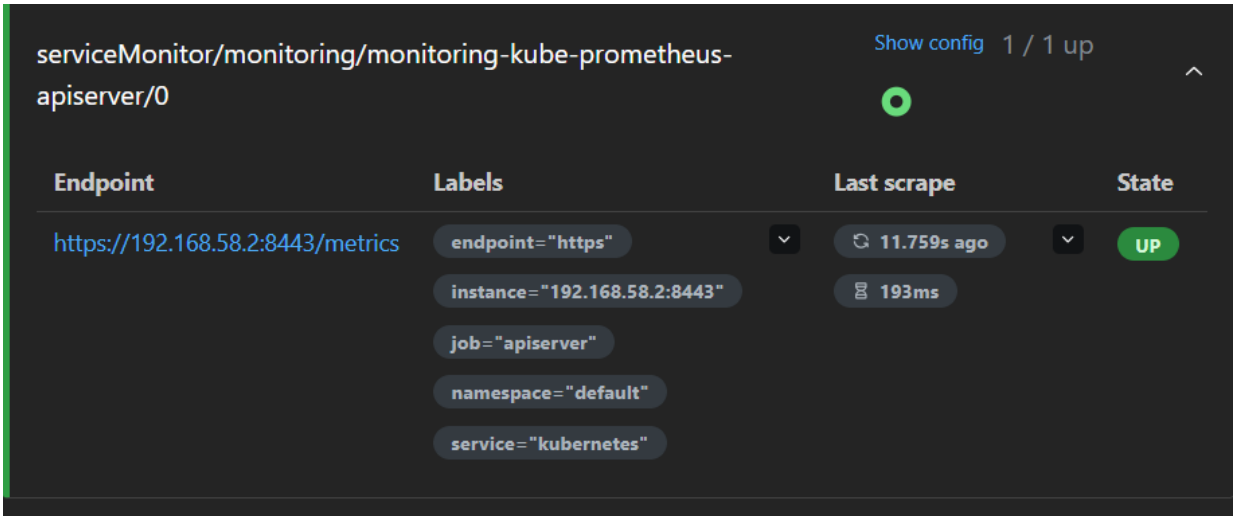

Evidence group: Observability | File: m9-prometheus-ui(2).png

## A38. Prometheus target for Grafana



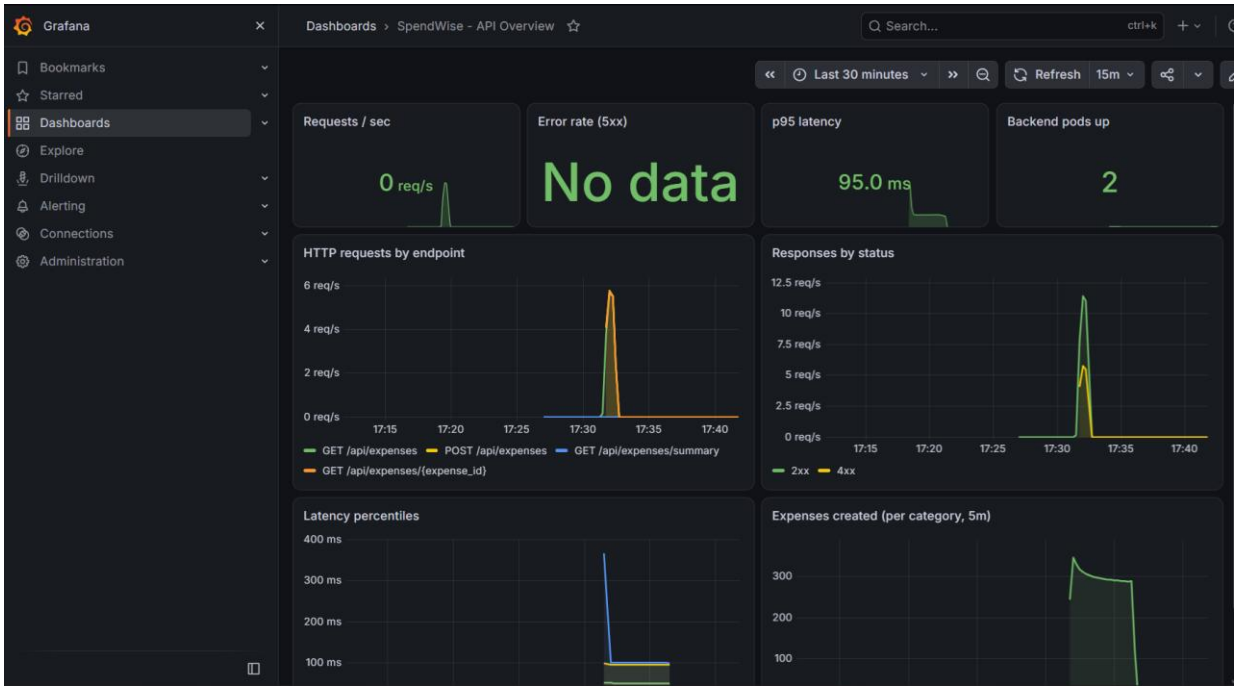

Evidence group: Observability | File: m9-prometheus-targets-grafana(2).png

## A39. Prometheus Kubernetes API server target



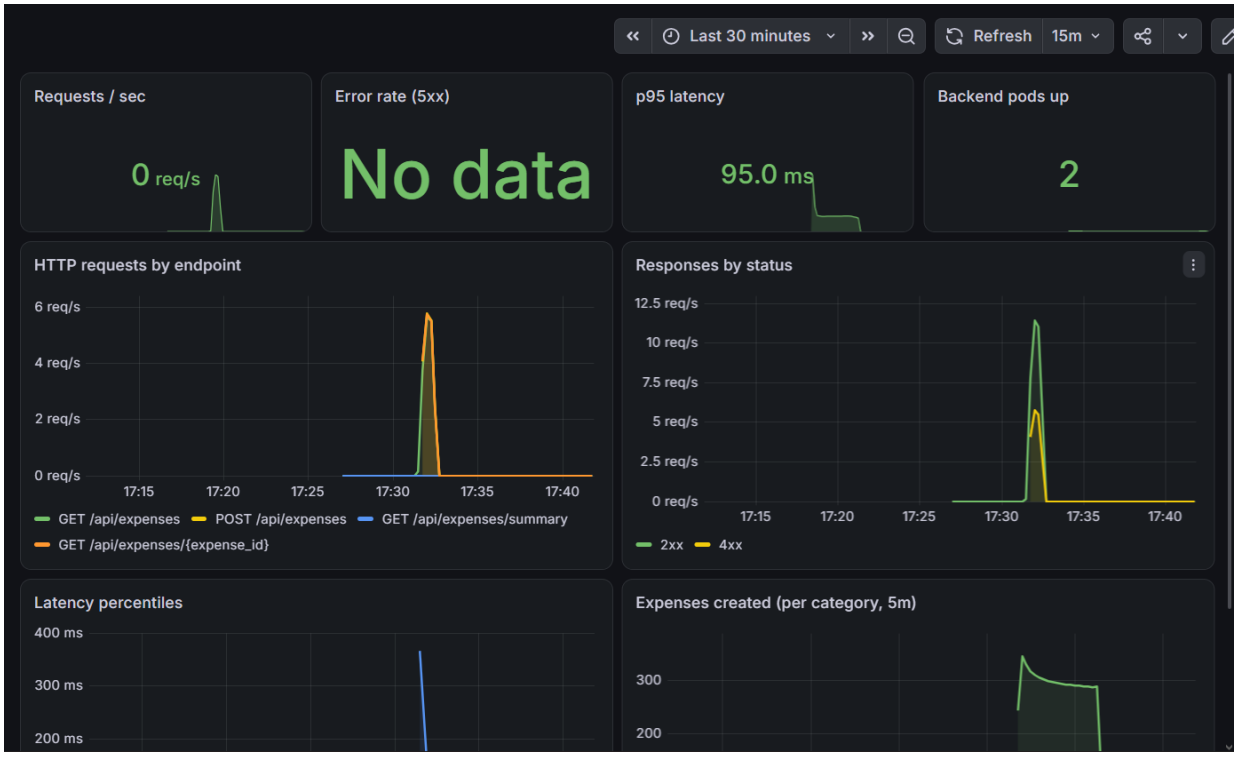

Evidence group: Observability | File: m9-prometheus-targets-apiserver.png

## A40. SpendWise API Overview dashboard



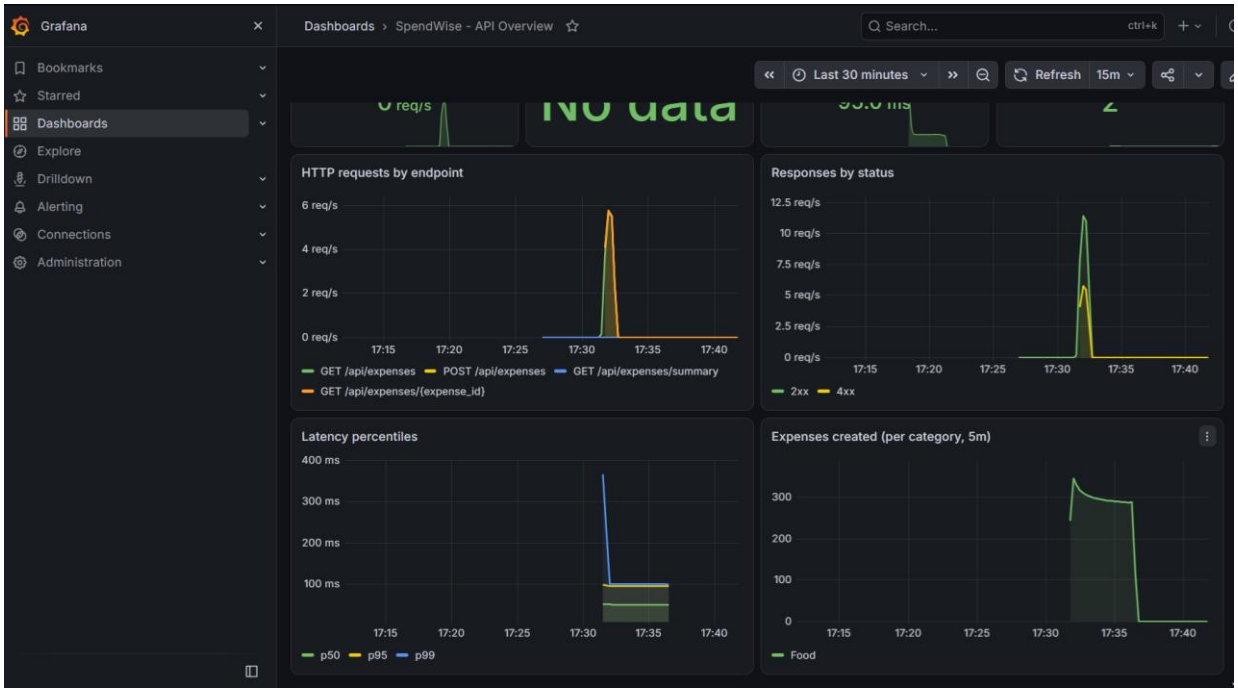

Evidence group: Observability | File: m9-grafana-dashboard(2).png

## A41. Grafana dashboard top-level indicators




Evidence group: Observability | File: m9-grafana-dashboard-top.png

## A42. Grafana dashboard detail panels




Evidence group: Observability | File: m9-grafana-dashboard-panels.png